

Energy Efficiency Report; Energy Audit of a Supermarket in Alaska

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Course Name

Energy Efficiency and Rational Use of Energy

Table of Contents

Introduction	3
Methodology	3
Step 1: Basic site information	4
Site characteristics	4
Energy needs of the reference building.....	8
Step 2: Equipment and Process Inventory	12
Step 3: Data Measurement	14
Step 4: Data Analysis.....	14
Consumption Analysis	14
Energy and Exergy Analysis of the Heating PSZ-AC.....	18
Envelope Energy Losses	24
Thermoeconomic Analysis for HVAC.....	25
Economic Analysis for Electricity and Gas Supply.....	27
Step 5: Improvement Proposal.....	29
Battery Energy Storage Optimization Plan	29
Heat pump replacement of gas-based HVAC system.....	33
Envelope improvement	39
Lightning Optimization Plan.....	44
Refrigeration Optimization Plan	47
Conclusion	48
Appendix	53

Introduction

This report presents a comprehensive energy audit and thermoeconomic analysis of a Reference Supermarket in Fairbanks, Alaska. The building analysed is part of a collection of reference buildings provided by the U.S. Department of Energy (DOE). This study aims to systematically analyse the supermarket's energy performance, identify the major energy-consuming components, and apply thermoeconomic methods to quantitatively measure the exergy destruction and corresponding costs. This analysis helps identify the optimization opportunities and apply the cost/energy-saving improvements.

Methodology

The overall methodology strictly follows the instructions of EEURE_CONTENT2A/B. Regarding energy audit, it employs a 5-step mechanism:

1. Basic information collection of the supermarket's characteristics and operations.
2. Equipment and process inventory (registration of energy-consuming equipment).
3. Data measurement.
4. Detailed data analysis, including exergy calculation and thermoeconomic analysis.
5. Proposal for the potential improvements.

In this study, all the data were generated using EnergyPlus 5.0 simulation software and DOE Reference Building models, incorporating Fairbanks' typical climate information.

The thermoeconomic analysis integrates exergy analysis and economic evaluations, focusing on the HVAC system, which is the main energy-consumption component. It involves:

1. Calculating the exergy flows and making the exergy balance of the HVAC system.
2. Assigning economic costs to exergy items to evaluate the economic implications of system losses.
3. Identifying the opportunities for improvements to the building's energy efficiency, and reducing costs.

Step 1: Basic site information

Site characteristics

The selected location to study is a supermarket in Fairbanks, Alaska, USA, at latitude of 64.82° and longitude of -147.9° . The building analysed is part of a collection of reference buildings provided by the U.S. Department of Energy (DOE) [1], modelling an average supermarket built in 2004 [2], providing a baseline of comparison and improvement for US's commercial structures. Fig.1 shows the layout of the location, which has 6 separate spaces, including the main sales area, a produce section, a dry storage space, a deli and bakery sections, as well as an office for the building's personnel. All of the areas are conditioned to a given temperature, and ventilated. The building has a total floor area of 4180.79 m^2 , 6.1m in floor to ceiling height, with one unique floor, and windows exclusively on the south facade (180° azimuth), the sales area, with a surface of 174.7 m^2 . Table 1 includes a detailed breakdown of the surface area for each section of the supermarket, as well as the walls and if they have a water supply.

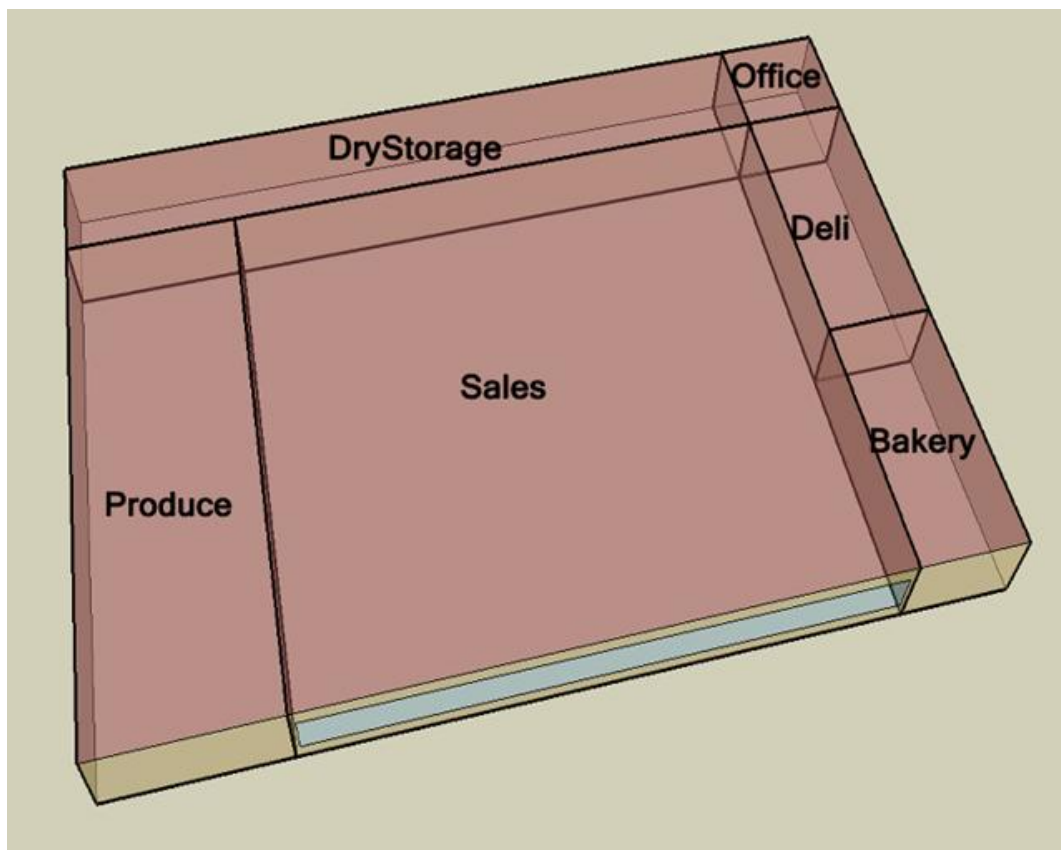


Figure 1: Floor plan of the supermarket, including labels for each subsection of the floor.

Table 1; Floor areas of all zones of the supermarket, perimeter data, and information on lights, gas supply and plugs.

Zone	Area (m2)	Volum e (m3)	Gross Wall Area (m2)	Water faucet (L/h serviced)	Lightin g (W/m2)	Plugs (W/m2)	Gas supply (W/m2)
Office	89	542	115	0	11.84	8.07	
Dry storage	622	3,792	477	0	8.61	8.07	
Deli	225	1,370	138	18.9	18.30	53.82	26.91
Sales	2325	14,177	324	0	18.30	5.38	
Produce	711	4,338	366	0	18.30	5.38	
Bakery	209	1,275	189	18.9	18.30	53.82	26.91
Perimeter Section		Construction	Gross Area (m2)	Azimuth (deg)	Tilt (deg)	Cardinal Directio n	
BAKERY_WALL_SOUTH		MASS NON-RES EXT WALL	60.34	180.00	90.00	S	
BAKERY_WALL__2		MASS NON-RES EXT WALL	128.73	90.00	90.00	E	
BAKERY_FLOOR		EXT-SLAB	209.04	0.00	180.00		
BAKERY_CEILING		IEAD NON-RES ROOF	209.04	180.00	0.00		
DELI_WALL_EAST		MASS NON-RES EXT WALL	138.39	90.00	90.00	E	
DELI_FLOOR		EXT-SLAB	224.72	0.00	180.00		
DELI_CEILING		IEAD NON-RES ROOF	224.72	180.00	0.00		
DRYSTORAGE_WALL_NORTH		MASS NON-RES EXT WALL	422.40	0.00	90.00	N	
DRYSTORAGE_WALL_WEST		MASS NON-RES EXT WALL	54.71	270.00	90.00	W	
DRYSTORAGE_FLOOR		EXT-SLAB	621.89	0.00	180.00		
DRYSTORAGE_CEILING		IEAD NON-RES ROOF	621.89	180.00	0.00		
OFFICE_WALL_EAST		MASS NON-RES EXT WALL	54.71	90.00	90.00	E	
OFFICE_WALL_NORTH		MASS NON-RES EXT WALL	60.34	0.00	90.00	N	
OFFICE_FLOOR		EXT-SLAB	88.84	0.00	180.00		
OFFICE_CEILING		IEAD NON-RES ROOF	88.84	180.00	0.00		
PRODUCE_WALL_SOUTH		MASS NON-RES EXT WALL	98.96	180.00	90.00	S	
PRODUCE_WALL_WEST		MASS NON-RES EXT WALL	267.12	270.00	90.00	W	
PRODUCE_FLOOR		EXT-SLAB	711.36	0.00	180.00		
PRODUCE_CEILING		IEAD NON-RES ROOF	711.36	180.00	0.00		
SALES_WALL_SOUTH		MASS NON-RES EXT WALL	323.44	180.00	90.00	S	
SALES_FLOOR		EXT-SLAB	2324.94	0.00	180.00		

Perimeter Section	Construction	Gross Area (m2)	Azimuth (deg)	Tilt (deg)	Cardinal Direction
SALES_CEILING	IEAD NON-RES ROOF	2324.94	180.00	0.00	

Table 2; Summary of wall and partition types, and their characteristics.

Partition type	R-value (m2 * K / W)	U factor (W /m2 * K)
Exterior walls	2.2	0.454
Roof	3.72	0.269
Window	-	2.615
Foundation	0.32	3.120

The building uses two energy sources, gas and electricity, and makes no use of district heating or cooling. For gas, the end uses are the general heating, water systems heating, and interior equipment (such as ovens or others). For electricity, end uses include cooling, refrigeration (broken down by medium and low temperature cooling), lighting, powering interior equipment (such as exhaust fans or socket appliances), and large fans. As there are no energy sources on-site, all electricity is purchased from the local utility, as well as the water.

There is a Package Single Zone Air Conditioner (PSZ-AC) for each area of the supermarket, with the characteristics of Table 3. The HVAC systems are designed to supply the maximum flow rate and can provide this same rated flow of Table 3 as outdoor air for ventilation, despite being designed for closed loop heating. The maximum flow rate of the fans is the same as the one mentioned in Table 3. Table 4 has information on the cooling characteristics of the Direct Expansion (DX) coils. Table 5 has information on the cooling sizing for the HVAC system and Table 6 has the same information for heating.

Table 3; Summary of Package Single Zone-AC for each location.

Zone	PSZ-AC number	Max flow rate (m3/s)	Min outdoor air flow rate (m3/s)	Cooling DX coil type	Gas heating coil nominal capacity (W)
Office	PSZ-AC 1	0.749306	0.047816	Single speed	20756.97
Dry storage	PSZ-AC 2 (+ economizer)	4.05	0.466417	Double speed	127768.86
Deli	PSZ-AC 3	1.61	0.337078	Double speed	157897.29
Sales	PSZ-AC 4 (+ economizer)	12.26	3.49	Double speed	543274.2
Produce	PSZ-AC 5 (+ economizer)	4.18	1.07	Double speed	176015.85
Bakery	PSZ-AC 6	1.82	0.313561	Double speed	65283.17

Table 4; Summary of DX coil characteristics for each location.

Item	Rated High Speed Air Flow Rate [m3/s]	Rated High Speed Total Cooling Capacity (gross) [W]	Rated High Speed Sensible Heat Ratio	Rated Low Speed Air Flow Rate [m3/s]	Rated Low Speed Total Cooling Capacity (gross) [W]	Rated Low Speed Sensible Heat Ratio
PSZ-AC:1	0.749306	12403.68	0.798655	-	-	-
PSZ-AC:2	4.05	66987.62	0.798655	1.35	22326.97	0.798655
PSZ-AC:3	1.61	26612.51	0.798655	0.535834	8869.95	0.798655
PSZ-AC:4	12.26	202903.10	0.798655	4.09	67627.60	0.798655
PSZ-AC:5	4.18	69159.29	0.798655	1.39	23050.79	0.798655
PSZ-AC:6	1.82	30127.73	0.798655	0.606611	10041.57	0.798655

Table 5; Summary of zone cooling design sizing for the HVAC system.

Zone	PSZ-AC number	Max flow rate (m3/s)	User design air flow rate (m3/s) (same as calculated cooling)	Design load (W)	Calculated design air flow (m3/s)
Office	PSZ-AC 1	0.749306	0.093	1120.95	0.077
Dry storage	PSZ-AC 2 (+ economizer)	4.05	0.466	1963.03	0.135
Deli	PSZ-AC 3	1.61	1.608	20784.69	1.340
Sales	PSZ-AC 4 (+ economizer)	12.26	3.487	0	0
Produce	PSZ-AC 5 (+ economizer)	4.18	1.067	0	0
Bakery	PSZ-AC 6	1.82	1.820	23380.48	1.517

Table 6; Summary of zone heating design sizing for the HVAC system.

Zone	PSZ-AC number	Max flow rate (m ³ /s)	User design air flow rate (m ³ /s) (same as calculated heating)	Design load (W)	Calculated design air flow (m ³ /s)
Office	PSZ-AC 1	0.749306	0.749	17598.62	0.624
Dry storage	PSZ-AC 2 (+ economizer)	4.05	4.047	95423.43	3.372
Deli	PSZ-AC 3	1.61	1.410	33239.20	1.175
Sales	PSZ-AC 4 (+ economizer)	12.26	12.257	290207.49	10.214
Produce	PSZ-AC 5 (+ economizer)	4.18	4.178	98709.05	3.428
Bakery	PSZ-AC 6	1.82	1.466	34515.66	1.221

The water system consists of a loop with maximum flow rate of 0.000011 [m³/s] or 0,011 [l/s]. The water heater has the same design flow rate, like the pump, which has a rated head of 0.1 pa. For the water heating there is a 0.4 m³ storage tank, with an 845 kW rating, 0.8 thermal efficiency, 1.72 recovery efficiency and an energy factor of 0.59.

Energy needs of the reference building

Regarding the location's conditions and requirements, the supermarket has general energy requirements that are almost the same of every type of such a building, and unique ones given by its location, with related challenges due to the extreme cold climate.

Regardless of the location, each supermarket needs to have heating, ventilation and air conditioning (HVAC), providing a comfortable temperature for customers and employees and ensuring good air quality. Moreover, being the aim of the facility to sell perishable goods, the presence of refrigerated display cases and walk-in freezers is essential. In presence of an on-site cuisine, hot food warmers, ovens, and grills for prepared meals are also required. Lightning systems are also present in both intern and outdoor spaces, with different design and specifics. In case of large windows, daylight utilization could decrease artificial lighting use. Furthermore, water and plumbing systems are present in the bakery and sales areas, together with a water heating system used by the deli.

Some of the aforementioned requirements are more demanding than in regular supermarkets, due to the extreme cold weather conditions in Fairbanks (Alaska), while other could take advantage of the low temperatures. Specifically, due to long winters with temperatures below -40°F (-40°C) [3], heating is a major energy cost.

The chosen supermarket consists of six areas: office, dry storage, deli, sales, produce, bakery. The zones are shown in Figure 1. Each zone has its specific functions, thus distinct requirements. The office area, where administrative and staff operations take place, require energy-efficient lighting, heating and cooling systems and ventilation to maintain air quality. In this zone, electronic equipment such as computers, monitors and printers would require energy supply. The dry storage area, which houses non-refrigerated goods, must have temperature control to prevent food's damage. The area also requires a performative ventilation system, to avoid the presence of potentially dangerous humidity that may spoil the goods. In the deli section, as in the bakery, where hot and cold foods are prepared and served, refrigeration, ovens, grills, and food warmers are ones of the most relevant energy consumers. Adequate ventilation is necessary to remove heat and fumes from the cooking hob, as to prevent humidity. The produce section, where fresh fruits and vegetables are stored, requires precise humidity control and refrigeration. Infact, produce is particularly vulnerable to freezing. Finally, the largest zone, the sales area, relies on lighting, heating and refrigeration to provide a suitable environment for customers.

Some building's energy needs are just constrained into the open-to-the-public hours, which are between 7 am to 10 pm (every day of the year, no closing day for holidays), while others are related to the products maintenance or security issues (lights on during night). The following graphs show a pattern of the schedules and consumptions during the 24 hours day.

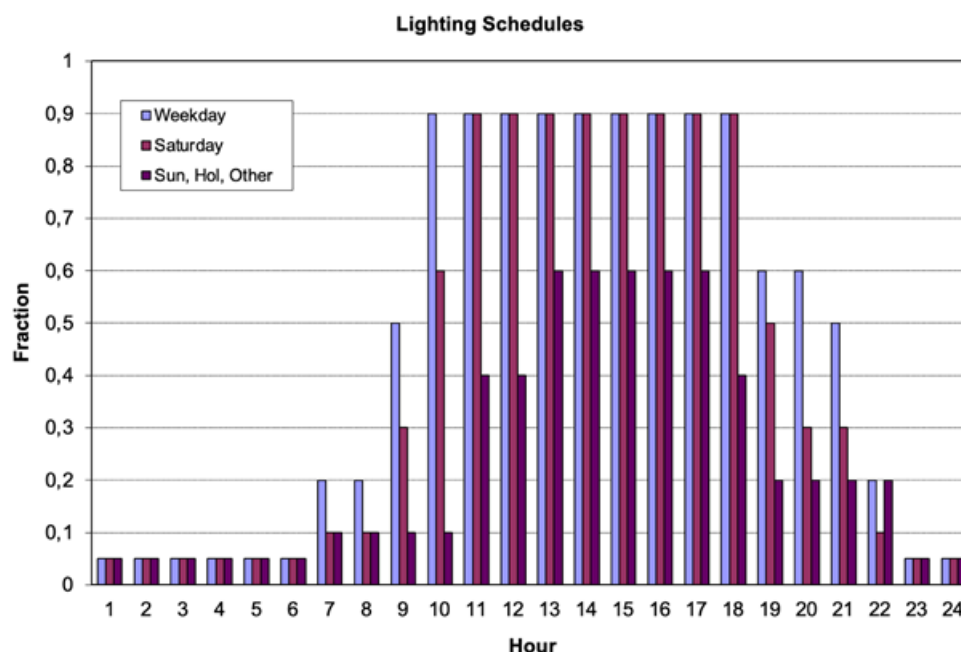


Figure 2; lightning schedules.

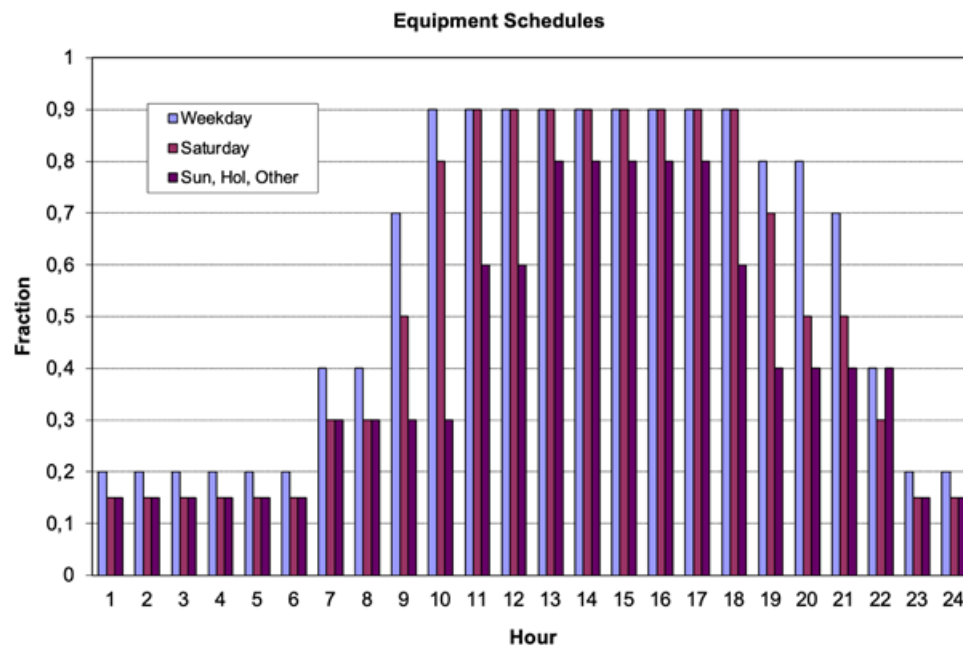


Figure 3; equipment schedules.

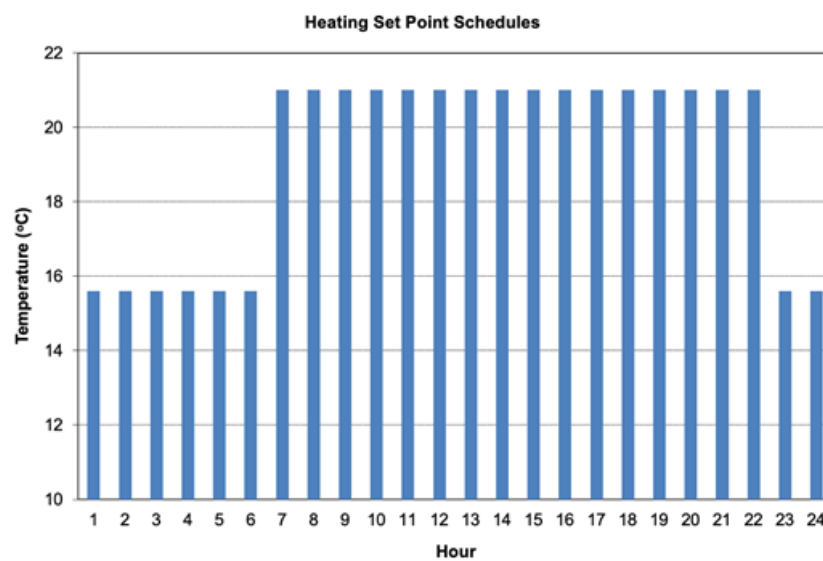


Figure 4; heating set point schedules.

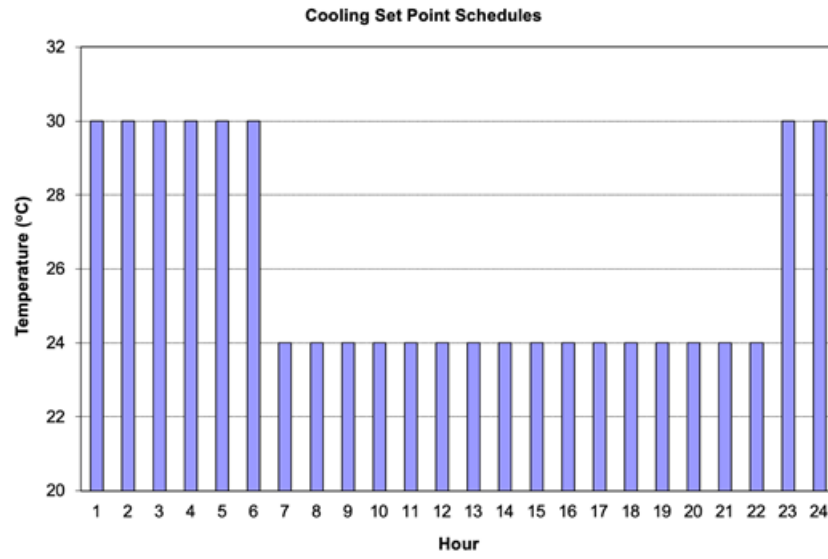


Figure 5; cooling set point schedules.

Regarding the energy sources that the building is using to fulfil its electrical consumption, gas and electricity are completely bought from the grid.

Step 2: Equipment and Process Inventory

The major energy-consuming equipment/systems are listed below:

1. HVAC Systems: Six PSZ-AC units each with gas-fired heating coils (coil efficiency range 78–80%) and electric DX cooling coils (COP 3.2–3.6), with 6911.11 GJ and 13.45 GJ total annual consumption respectively.
2. Lighting: Predominantly fluorescent lighting (fluorescent type is the standard model type of lighting for DOE reference building model), totalling approximately 1145.42 GJ annually.
3. Refrigeration appliances: Significant energy consumer at 2,824 GJ/year.
4. Fans: Essential for air circulation and ventilation, consuming about 975 GJ/year.
5. Miscellaneous Equipment: Electric plug loads amounting to 785 GJ, and miscellaneous gas equipment at 199 GJ/year.
6. Hot Water System: Natural gas-based water heating system consuming approximately 30 GJ annually.

The following part gives a more detailed specification of some major equipment/systems in terms of their efficiencies:

PSZ-ACs: Cooling Coils

Table 7; PSZ-ACs Cooling COPs.

Number	Type	COP
PSZ-AC 1	Single Speed	3.67
PSZ-AC 2	Two Speed	3.30
PSZ-AC 3	Two Speed	3.50
PSZ-AC 4	Two Speed	3.23
PSZ-AC 5	Two Speed	3.30
PSZ-AC 6	Two Speed	3.50

The Cooling Coil with Single Speed has only fixed speed (100% capacity) and the system is either ON or OFF. On the other hand, the coil with two speeds has the possibility to switch from the high speed (100%) to the medium/low speed (to be fixed). This second type is more efficient since it has fewer ON/OFF cycles thus less power usage, better temperature control and also it is relatively quieter (since the low speed produces less noise).

Some other figures relative to the Single Speed coil are available:

Table 8; Single speed coil Efficiencies.

Type	EER in SI Units [W/W]	EER in IP Units [Btu/W-h]	IEER in SI Units [W/W]	IEER in IP Units [Btu/W-h]	SEER in SI Units [W/W]	SEER in IP Units [Btu/W-h]
Single Speed Coil	2.98	10.18	3.19	10.89	3.42	11.65

PSZ-ACs: Heating Coils

Table 9; PSZ-ACs Heating Efficiencies.

Number	Type	Coil Nominal Efficiency
PSZ-AC 1	Gas	0.80
PSZ-AC 2	Gas	0.78
PSZ-AC 3	Gas	0.78
PSZ-AC 4	Gas	0.78
PSZ-AC 5	Gas	0.78
PSZ-AC 6	Gas	0.80

Fans

In addition to the Fans comprehended in the PSZ-AC there are three supplementary Fans in the building: one in the Bakery, one in the Deli and one in the Sales zone. In total the Fans and their efficiency ratings are:

Table 10; Fans Rated Power and Efficiencies.

Number	Type	Rated Power	Total Efficiency
Bakery Exhaust Fan	Zone Exhaust	130.92	0.34
Deli Exhaust Fan	Zone Exhaust	0	1
Sales Exhaust Fan	Zone Exhaust	0	1
PSZ-AC 1	Constant Volume	853.61	0.55
PSZ-AC 2	Constant Volume	7591.61	0.59
PSZ-AC 3	Constant Volume	1758.18	0.57
PSZ-AC 4	Constant Volume	20767.58	0.60
PSZ-AC 5	Constant Volume	7837.72	0.59
PSZ-AC 6	Constant Volume	1990.42	0.57

Pumps

In the building there is a pump used in the water system, with the following the characteristics:

Table 11; Pump characteristics.

Number	Control	Type	Power [W]	Head [pa]	Motor Efficiency
SWHSYS 1 PUMP	Intermittent	Variable Speed	0.00	0.10	1.00

Service Water Heating

Finally, there is also a Heater system for the water:

Table 12; Service Water Heating characteristics.

Number	Type	Thermal Efficiency	Recovery Efficiency	Energy Factor
SWHSYS 1 HEATER	Mixed	0.80	1.72	0.59

Step 3: Data Measurement

As mentioned above, all the data were derived by EnergyPlus 5.0 simulation software, based on DOE Reference Building models [2]. The simulation incorporates Fairbanks' typical climate information to ensure the realistic energy performance evaluations.

The data measurements used are based on a specific design day; for example, the zone heating data is derived from a design day called FAIRBANKS ANN HTG 99.6% CONDNS DB (1/21 07:00:00). This day is used as a reference point to evaluate the operational conditions of the HVAC system in heating mode. A similar reference day is specified for cooling needs.

Step 4: Data Analysis

Consumption Analysis

The following series of graphs provides a comprehensive analysis of energy and resource consumption patterns based on the detailed monthly data.

Figure 6 summarizes the annual consumption of electricity (in GJ) and natural gas (in GJ) across different categories such as heating, cooling, interior lighting, exterior lighting, interior equipment, fans, water systems and refrigeration. Notably, natural gas consumption is heavily driven by heating while refrigeration (specific to fridges and

freezers) is the main contributor to electricity consumption. In addition, fans, interior equipment and interior lighting also show significant electricity consumption. Other categories like cooling, exterior lighting and water systems consume minimal energy. Regarding the overall share of electricity and natural gas in total energy consumption, the graph below shows that natural gas accounts for 55.4% (7,141 GJ), while electricity accounts for 44.6% (5,744 GJ) which indicates that natural gas is the primary energy source in this building.

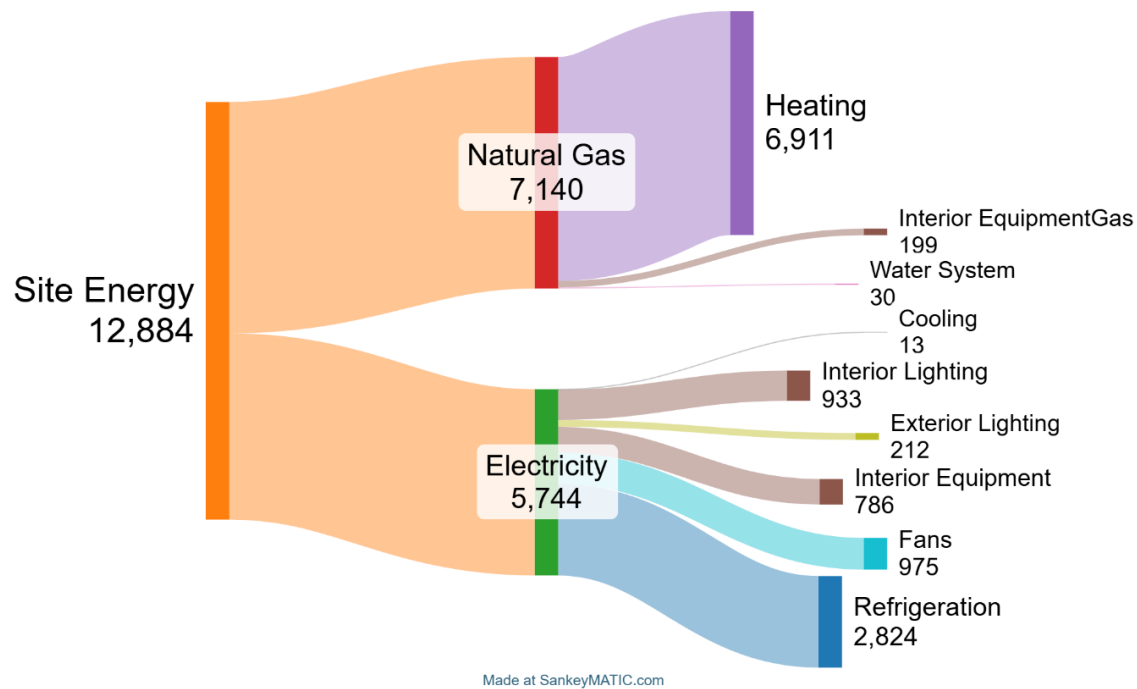


Figure 6; Annual consumption summary (all the numbers are expressed in GJ).

Figure 7 illustrates the monthly carbon equivalent emission (CO₂e in kg) for the facility. CO₂e is a metric used to aggregate emissions of different greenhouse gases into a single value, based on their relative contribution to global warming [4]. One remarkable observation is about the seasonal trend that emissions are the highest in winter months, particularly in January (58,713 kg) and December (57,570 kg) while lower in summer months, with the lowest value in June (41,359 kg). It's most likely due to the variable heating demand for different seasons.

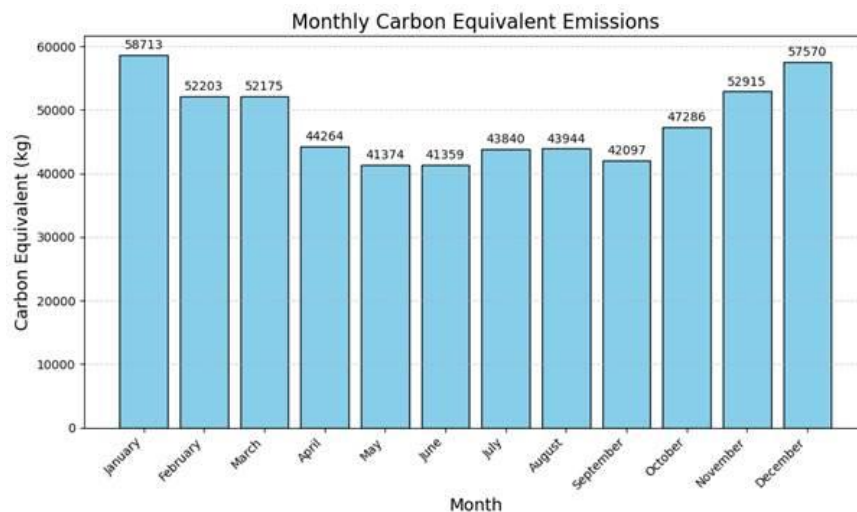


Figure 7; Monthly carbon equivalent emissions.

Figure 8 indicates the monthly average and maximum electricity load (in W). Notably, the peak load is in Summer, probably driven by refrigeration systems. In addition, the average electricity demand remains relatively stable, ranging between 105 kW and 155 kW. This implies that the cooling demand for this building is relatively low.

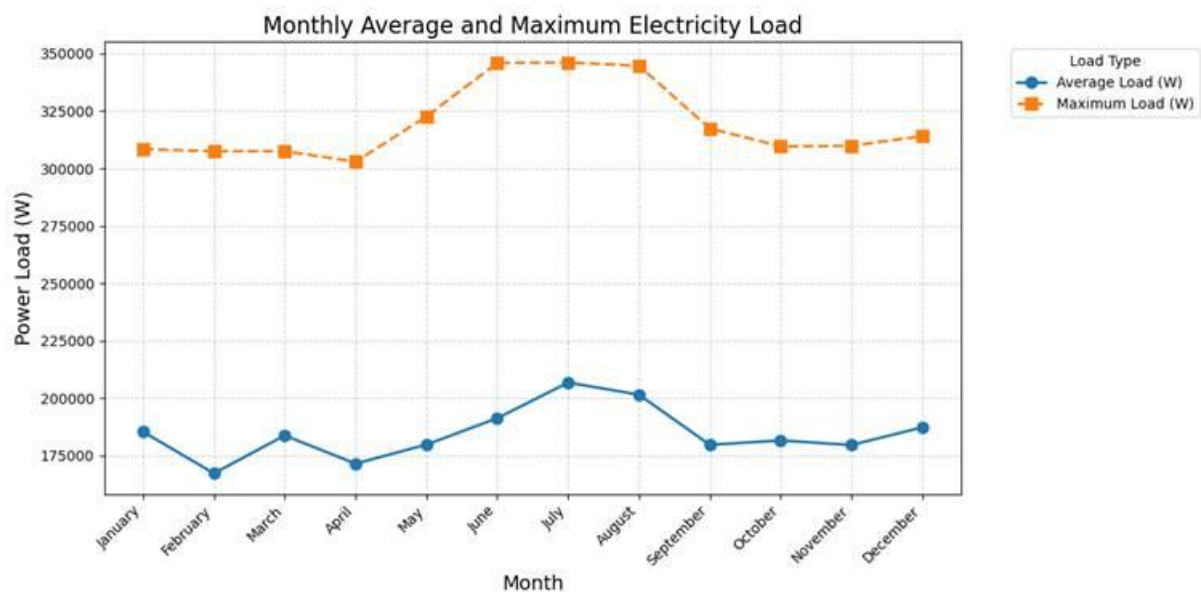


Figure 8; Monthly average and maximum electricity load.

Figure 9 displays the monthly average and maximum heating load (in W) across all HVAC units. The maximum heating load is the sum of the 6 HVAC units peak load in each month, so it will be higher than the actual peak load. However, from this graph, it's remarkable that the trend of the average heating load is identical with the trend of the carbon emission that peaks in winter and drops in summer. Meanwhile, there is a significant difference between the average and maximum heating load. The maximum heating loads are much higher than the maximum heating loads, in accordance with the previous

observation that gas, associated with heating, is the main energy source for the supermarket.

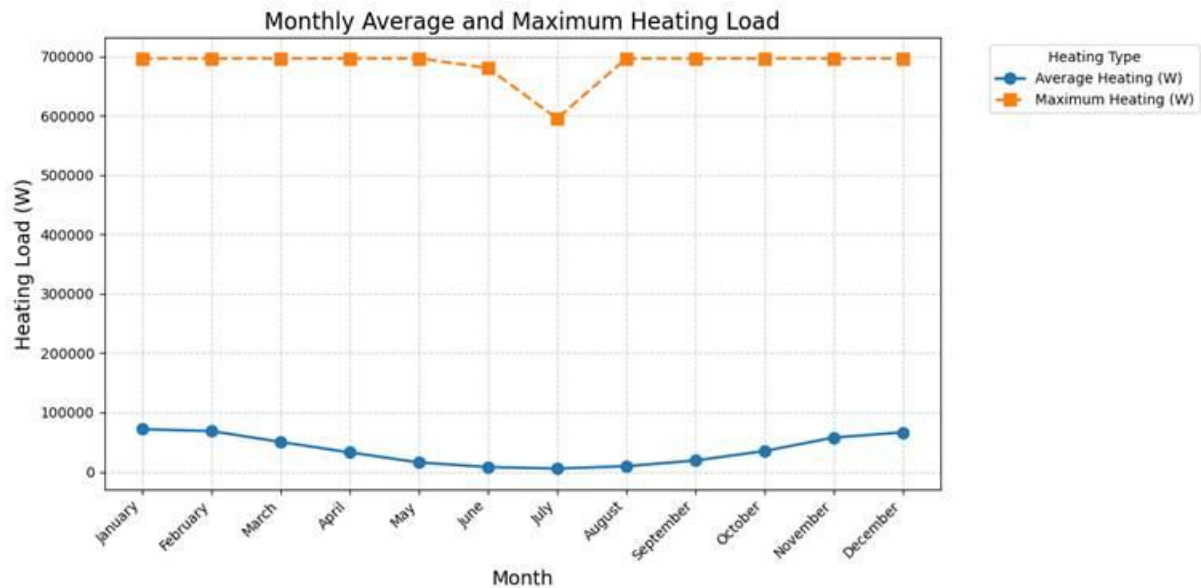


Figure 9; Monthly average and maximum heating load.

Accordingly, Figure 10 shows the monthly average and maximum cooling demand (in W) across all HVAC units. It peaks in summer, with the minimal demand in winter, and it is considerably smaller than heating loads.

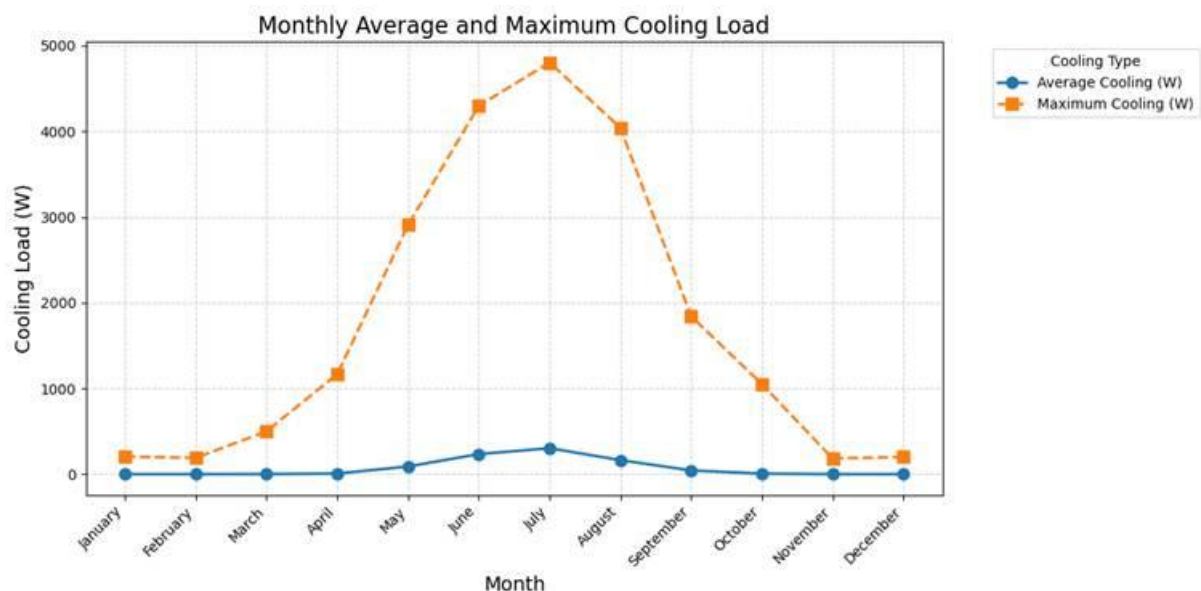


Figure 10; Monthly average and maximum cooling load.

In addition, as mentioned above, the building consumes **12,884.67 GJ** of energy annually, while the total source energy required from the production site is **28,315.35 GJ**. The site-to-source energy conversion factors are:

Table 13; Site to Source Conversion Factor.

	Site --> Source Conversion Factor	Efficiency
Electricity	3.572	27.99%
Natural Gas	1.092	91.57%

Considering that the building consumes 5744.07 GJ of electricity and 7140.60 GJ of Natural Gas, it is possible to calculate the average efficiency from Source to Site:

$$\text{Total Source to Site Efficiency} = \frac{12884.67}{28315.35} = 0.4550 = 45.5\%$$

$$\text{Total Site to Source Conversion Factor} = (0.4550)^{-1} = 2.197$$

Energy and Exergy Analysis of the Heating PSZ-AC

The exergy audit is a framework for evaluating energy systems by quantifying the energy and exergy waste relative to the energy and exergy supplied, both at the system-wide or component levels. It also provides a thermodynamic measure of energy quality or usefulness, allowing a better understanding of how effectively energy can be converted into work. Additionally, exergy serves as a basis to define efficiencies, allowing more accurate assessments of system performance and thus potential improvements. For the studied case, from an energy consumption point of view, the most demanding utilities are the interior equipment, and heating from the HVAC systems. These are followed by fans and lightning. While for the interior equipment and lightning an exergy analysis is not possible, due to the lack of process input and output data, the exergy analysis is possible for the optimization of the HVAC system. The energy analysis of section 6 indicates that heating energy needs are much larger than cooling, since the former represents 99.8% of all energy needs of the HVAC system. Consequently, the following analysis will focus on the HVAC heating operations, as it represents the main energy consumption category, and the HVAC cooling operations will not be analysed in depth due to their low impact.

The heating system is designed as shown in the following scheme, which has the same scheme for all six zones. Both the heating and cooling system are designed work in closed loop as a default option, meaning that the air to be warmed or cooled is taken from the inside are of the supermarket. This way allows a huge save in energy, particularly in the gas consumptions that does not have to warm up the air from the environmental conditions (-42.5 °C) to the desired temperature (40 °C).

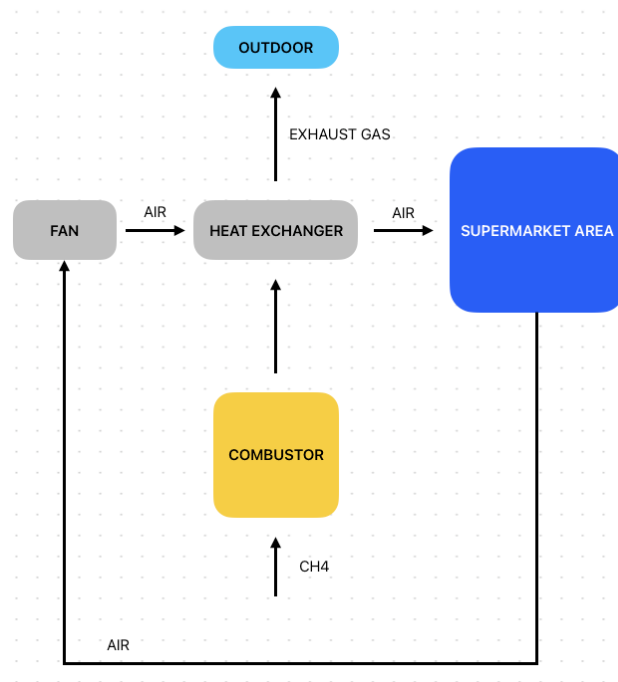


Figure 11; Heating system.

The cooling system, instead, is simpler due to the absence of the gas combustor and heat exchanger, being powered by electricity rather than gas (CH4).

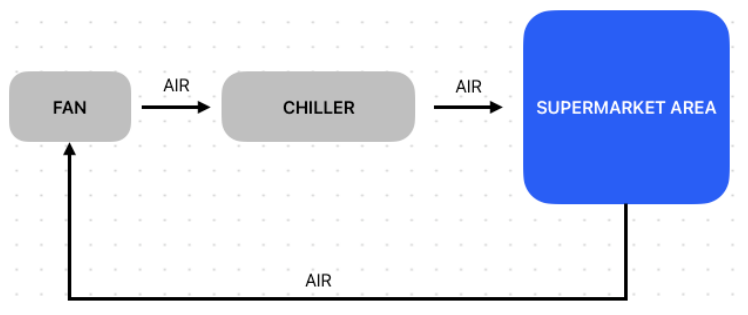


Figure 12; Cooling system.

The energy analysis was conducted by evaluating the input energy of the AC system, including fan power and heating power, and the output energy transferred to the air as it passes through the heating system. Specifically, the following formula was used:

$$W_{out} = \dot{m} * c_p * (T_{out} - T_{env})$$

Where T_{env} is the room temperature of the supermarket, given the closed loop cycle, and

$$W_{in} = FanPower + HeatingPower$$

With this simple calculation, it is possible to determine the energy efficiency of the six different heating systems. The results are shown in Table 14.

Table 14; PSZ-ACs Heating Energy Efficiencies.

AC system	W_in (kW)	W_out (kW)	Efficiency (%)
PSZ-AC 1	22.85	18.51	81.04
PSZ-AC 2	129.92	99.99	76.96
PSZ-AC 3	44.37	34.81	78.46
PSZ-AC 4	392.82	302.71	77.06
PSZ-AC 5	134.38	103.21	76.80
PSZ-AC 6	45.13	36.19	80.20

These values are consistent with typical performance levels for AC heating systems and highlight the potential to increase energy efficiency by using more advanced equipment. However, the exergy analysis provides a more insightful perspective, revealing the true potential for improvement in system performance. The calculated exergy values can be found in the Appendix.

During the study, different data and assumptions were used and made:

- The air flows through a closed loop system
- $c_{p_{air}} = 1.005 \frac{kJ}{kgK}$ [5]
- $density (\rho_{air}) = 1.293 \frac{kg}{m^3}$ [6]
- $R_{air} = 0.287 \frac{kJ}{kgK}$ [7]
- Gas to be burned is CH_4 ; $LHV_{CH_4} = 49,800 \frac{kJ}{kg}$ [8]
- Pressure CH_4 (p_{CH_4}) = 12 bar
- Molar mass CH_4 ($\bar{M}_{CH_4}$) = $16.043 \frac{g}{mol}$ [9]
- Chemical exergy CH_4 ($b_{ch_{CH_4}}$) = $51406.84 \frac{kJ}{kg}$
- Physical exergy CH_4 ($b_{ph_{CH_4}}$) = $381.95 \frac{kJ}{kg}$

The CH_4 's chemical exergy is evaluated according to the following equations in the specific case:

$$\bar{b}_{CH_4}(T_0, p_0) = \left[\bar{g}_{CH_4} + 2 * \bar{g}_{CO_2} - \bar{g}_{CO_2} - 2 * \bar{g}_{H_2O_{liq}} \right] (T_0, p_0) + \left[\bar{b}_{ch,CO_2} + 2 * \bar{b}_{ch,H_2O,liq} - 2 * \bar{b}_{ch,CO_2} \right] = 824720 \left[\frac{J}{mol} \right],$$

where $\bar{b}$ and $\bar{g}$ are values in molar basis.

$$b_{ch,CH_4}(T_0, p_0) = \frac{\bar{b}_{CH_4}(T_0, p_0)}{\bar{M}_{CH_4}} = 51406.84 \left[\frac{kJ}{kg} \right]$$

To calculate the exergy, the following equation is being utilized:

$$b_{ph}(T, p) = c_p * (T_g - T_0 - T_0 * \ln\left(\frac{T_g}{T_0}\right) + R_g * T_0 * \ln\left(\frac{p_g}{p_0}\right)$$

where g is related to the gas (CH_4) or air and 0 refers to the dead state environmental conditions.

Each area of the supermarket has its own characteristics: the relevant ones for the exergy's calculation are the temperature of the environment, of the air or gas, the pressure rise after the fan, and the fan's power consumptions. All these data are provided by the U.S. Department of Energy (DOE) and are shown in the tables in the Appendix, together with the exergy values. The "C" indicates that the values were calculated, otherwise they were already provided.

The graphs below display the exergy flow of the six heating systems located in the supermarket. It can be said without any doubt that every PSZ-AC system predominantly relies on Natural Gas Chemical Exergy, whereas the contributions made by Physical Exergy and Fan Electricity Exergy are considerably lower. When looking at the heating system for the sales area, having the highest total energy use, it is possible to see that from the 406.12 kW input, only 5.65 % is actually converted into Warm Air Physical Exergy. The remaining 383.16 kW are lost, primarily due to inefficiencies in the system. This trend is found to be quite uniform for all six air conditioners, with each running at about 5% efficiency. This clearly indicates a big opportunity for exergy loss reduction and, in turn, lower operating cost by applying a more efficient heating system. Besides, heating is responsible for more than half of the energy consumed by the supermarket (6911 GJ), as shown in Figure 6, indicating a large potential impact of improvements regarding the heating systems.

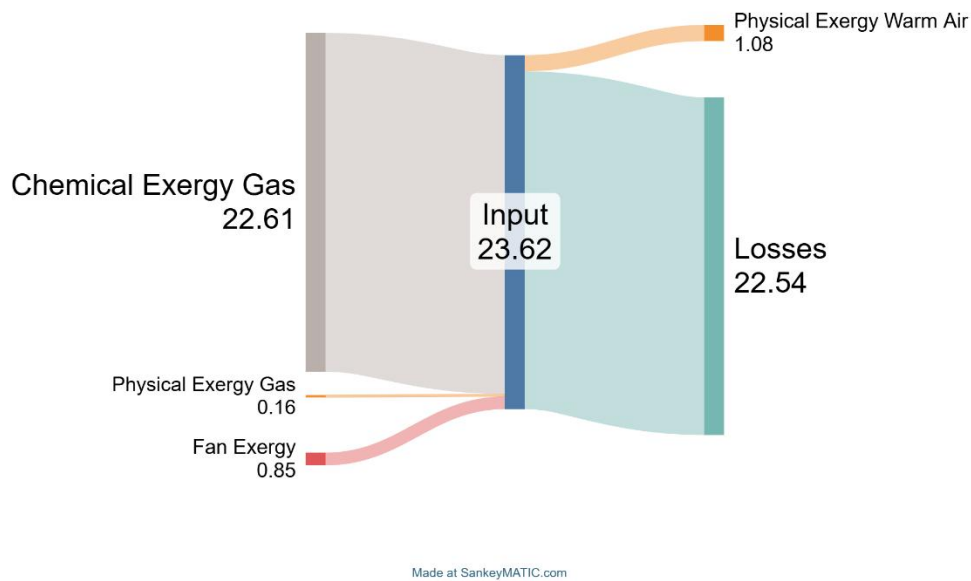


Figure 13; PSZ-AC 1, Exergy Scheme, all the values are expressed in kW.

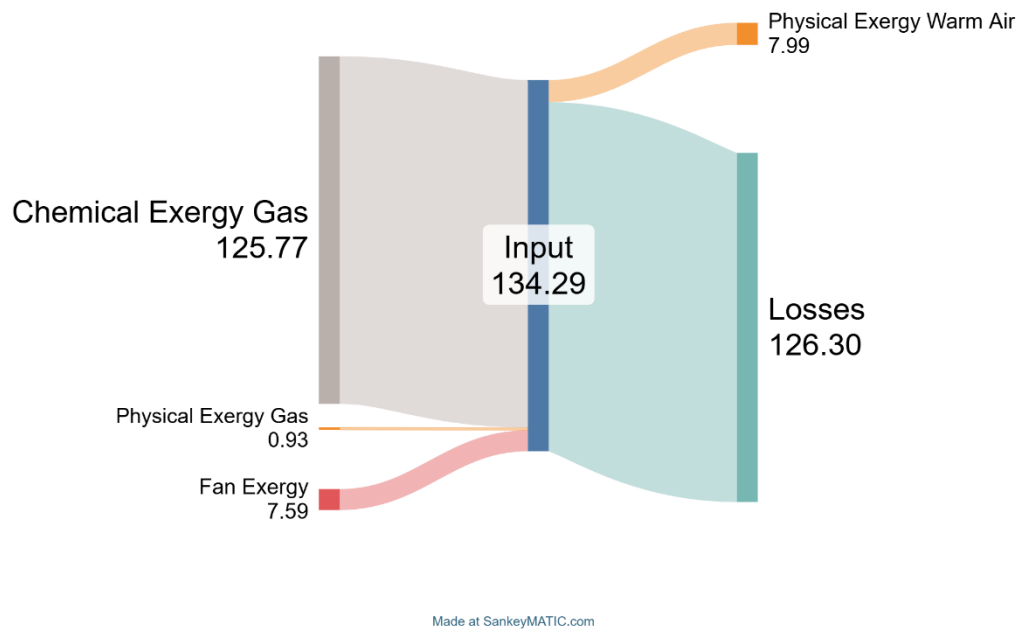


Figure 14; PSZ-AC 2, Exergy Scheme, all the values are expressed in kW.

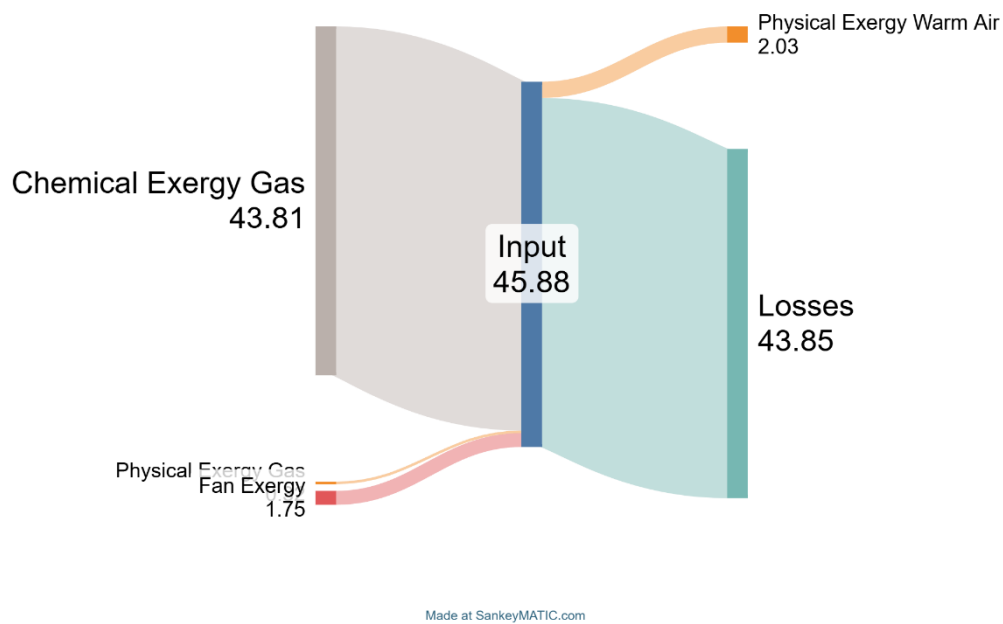


Figure 15; PSZ-AC 3, Exergy Scheme, all the values are expressed in kW.

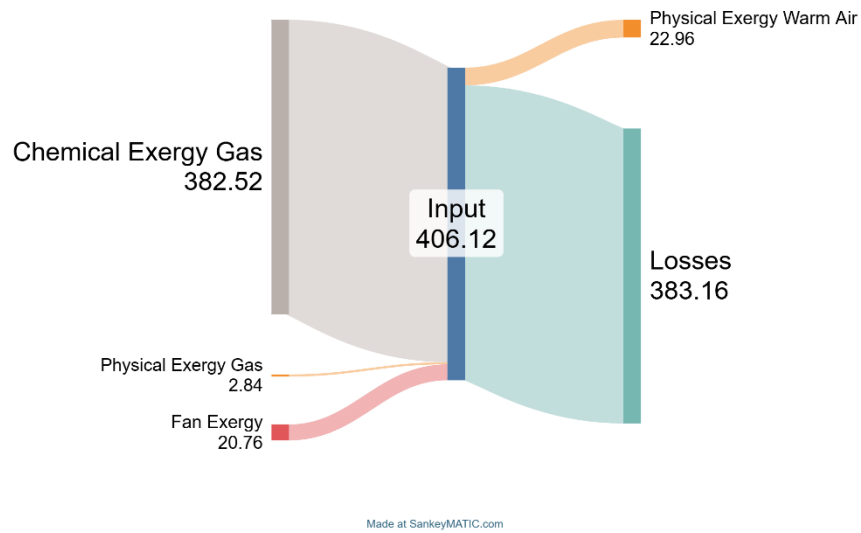


Figure 16; PSZ-AC 4, Exergy Scheme, all the values are expressed in kW.

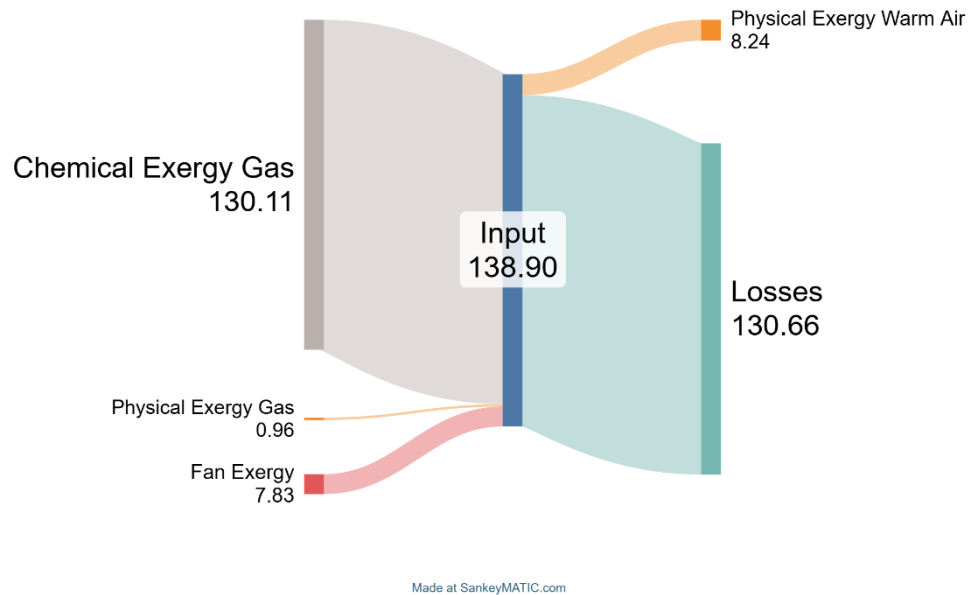


Figure 17; PSZ-AC 5, Exergy Scheme, all the values are expressed in kW.

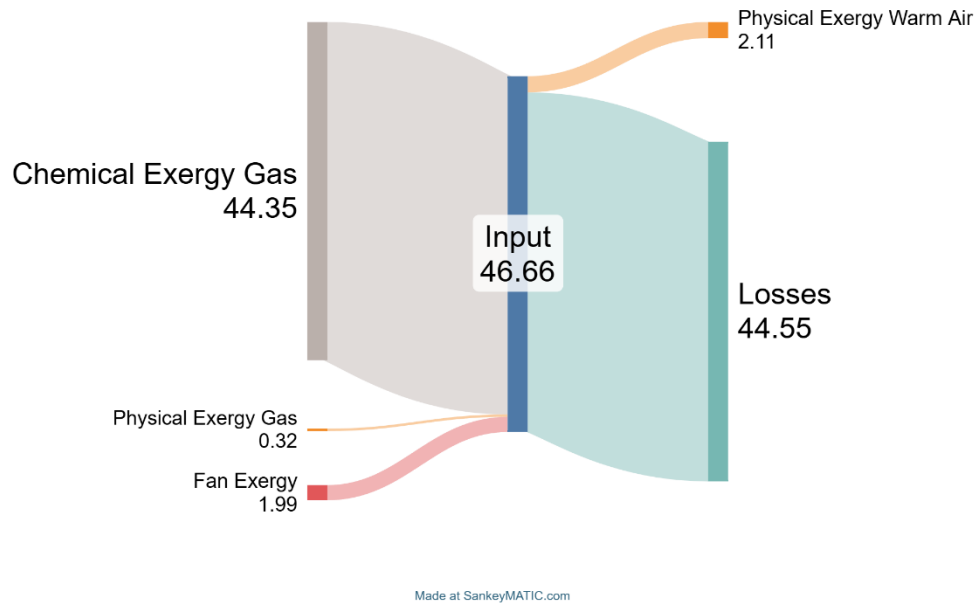


Figure 18; PSZ-AC 6, Exergy Scheme, all the values are expressed in kW.

From this analysis it is possible to say that in the normal course of events, the HVAC system is observed to register high energetic efficiency but low exergetic efficiency, especially in thermal systems where most of the supplied energy becomes low-quality heat [10]. This emphasizes why exergy analysis is crucial for gaining a realistic picture of the true performance and sustenance of the system.

Envelope Energy Losses

In the process of analysing heating needs for the supermarket, the losses through the building envelope, namely the walls, roof, floor and windows, are a considerable factor influencing the building's energy efficiency. Given the temperature difference between the target room temperature of 21°C, the environment at -42.5°C and the soil temperature at -3°C [11] during the reference day for heating needs, heat losses represent a large expense. The building's envelope is characterized by the respective U values (W/m² K) for the wall, window, roof and foundation, which are 0.454, 2.615, 0.269 and 3.120 respectively. The U-values for the wall, roof and floor are above the maximum allowed for new constructions in most EU countries [12] and in the USA [13], which stands between 0.11-0.18 W/m² K, suggesting there is ample room for improvement. The corresponding heat losses Q (W) can be calculated according to the heat transfer formula:

$$Q = A * U * \Delta T$$

Where A represents the surface area in m² of the respective envelope section, and ΔT is the temperature difference between either side of the wall, window, floor or roof. Table

15 presents an overview of the heat losses for each respective section, as well as those for the entire building. Additionally, the percentage of heat losses corresponding to either the wall and window surfaces, the floor or the roof are also analysed, suggesting that an insufficient floor insulation is leading to the largest heat losses in the building.

Table 15: Supermarket section breakdown of wall, window and roof surface exposed to the exterior, the respective heat losses and percentage of total losses from each category.

Building Zone	Roof Area (m ²)	Gross Wall Area (m ²)	Window Glass Area (m ²)	Loss floor (%)	Loss windows and wall (%)	Loss roof (%)	Total heat loss (kW)
Office	89	115	0	0.579	0.289	0.132	11.488
Dry Storage	622	477	0	0.656	0.194	0.150	70.948
Deli	225	138	0	0.682	0.162	0.156	24.656
Sales	2325	324	175	0.690	0.152	0.158	252.141
Produce	711	366	0	0.701	0.139	0.160	75.974
Bakery	209	189	0	0.634	0.221	0.145	24.676
Building total	4181	1610	175	0.657	0.193	0.150	459.884

Thermoeconomic Analysis for HVAC

With the corresponding exergy balances for the HVAC systems of the building, a thermoeconomic analysis can be conducted for the heating process in each designated area. Across all HVAC systems, the exergy flows can be grouped into the scheme shown below in Fig.19. The exergy destroyed during the process and the exergy dumped to the environment can be grouped into one single term, given that both processes result in exergy not being used for its purpose of heating up the air. Attributing a cost C_i to each exergy flow, and including the annual maintenance and operation cost of the HVAC system Z_{hvac} , the following balance can be obtained;

$$-C_{gas} * E_{gas} - C_{elec} * E_{wfan} + C_d * E_{destroyed} + C_{airout} * E_{airout} - C_{airin} * E_{airin} = Z_{hvac} .$$

The balance can be simplified if we know that the inlet air has a 0 exergy value, since we disregard the chemical exergy of the air during the process (since it suffers no chemical transformations), and the dead state conditions are taken as those of the inlet air. Furthermore, the C_d term will be assumed zero, for which we obtain the simplified expression

$$-C_{gas} * E_{gas} - C_{elec} * E_{wfan} + C_{airout} * E_{airout} = Z_{hvac} .$$

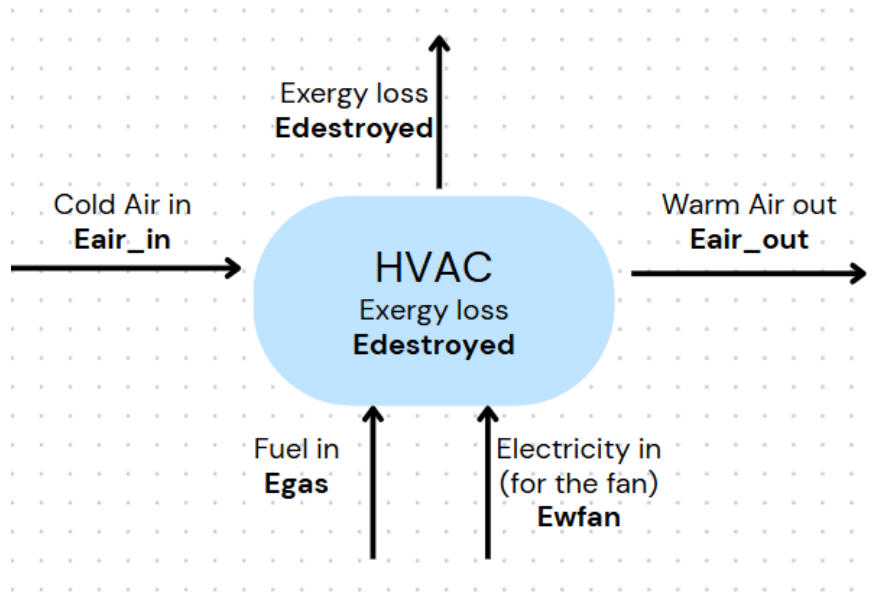


Figure 19; Basic exergy flows of the HVAC system.

The costs for the gas and electricity inputs are known from the supermarket's reference database, listed as 0.004151 \$/MJ for C_{gas} and 0.088521 \$/kWh for C_{elec} . A value for the annual maintenance and operational costs associated with HVAC systems can be obtained from the reference database provided by the *American Society of Heating, Refrigerating And A-C Engineers* (ASHRAE)[43] which suggests a typical value of 0.287 \$/sf or 3.089 \$/m². Knowing the surface area of each section, Z_{hvac} can be estimated for each system, from which the only remaining unknown C_{air_out} can be found for each sector of the supermarket. To work conveniently, costs and exergy flows are converted to an hourly basis, \$/hour and kWh respectively. The terms can be grouped into

$$CP = C_{air_out} * E_{air_out}$$

for the product, which is the warm air, and

$$CF = C_{gas} * E_{gas} + C_{elec} * E_{wfan}$$

for the costs per exergy flow of the fuels or input energy to the system, such that a further simplified balance of the system (when C_d are assumed zero) becomes

$$Z = -CF + CP \quad Z = -CF + CP .$$

Finally, thermoeconomic factors can be derived to inquire on the influence of different factors on the overall costs of the system. Table 16 lists all the relevant values for each HVAC system, as well as the thermoeconomic factors for the heating process.

Table 16; Summary of unit costs and thermoeconomic factors of HVAC systems in heating mode for the respective sectors of the supermarket.

	Zhvac operation and maintenance HVAC cost (\$/h)	Ewfan (kWh)	CF fan (\$/h)	Egas (kWh)	CF gas (\$/h)	Eair_out (kWh)	CP (\$/h)	Fz (Operation and maintenance ratio)	Ff (fuel ratio)	CF gas /CF (%)
office	0.000522162	0.85	0.075	22.78	0.3405	1.0828	0.416	0.0013	0.9987	81.90
dry storage	0.003655194	7.591	0.671	126.71	1.893	7.9915	2.569	0.0014	0.9986	73.81
deli	0.001320804	1.758	0.155	44.13	0.659	2.0356	0.816	0.0016	0.9984	80.91
sales	0.013664966	20.767	1.838	385.36	5.758	22.9685	7.610	0.0018	0.9982	75.80
produce	0.004181058	7.837	0.693	131.08	1.958	8.248	2.656	0.0016	0.9984	73.84
bakery	0.001228644	1.99	0.176	44.69	0.667	2.1165	0.845	0.0015	0.9985	79.13

As evidenced by the thermoeconomic factors Fz and Ff, the main cost sources are the fuel sources for the HVAC systems, which comprise the gas and electricity inputs. Consequently, the main optimization efforts need to be geared towards fuel savings. The last value of Table 10 suggests that gas consumption makes up from 74% to 82% of the total fuel costs CF, therefore particular focus needs to be placed on the heat input from the gas as the primary cost source of HVAC systems in heating mode, which in turn make up over half of the facility's energy use.

Economic Analysis for Electricity and Gas Supply

According to the dataset, the annual cost for electricity (Average Annual Rate: 0.08852 (\$/kWh)) and gas (Average Annual Rate: 0.00415 (\$/MJ)) is 141,241.79\$ and 29,638.78\$ respectively, so that the total cost is 170,880.57\$ (in 2010). This suggests that whilst gas consumption is the largest energy source, it is a lot cheaper than electricity, due to the price difference, for which the potential economic savings from gas consumption reduction will be less significant than electricity related savings.

Figure 20 breaks down the monthly electricity utility charges into energy charges, demand charges, service charges, and taxes. Energy charges are the costs associated with the actual consumption of electrical energy [14]. Demand charges are based on the highest rate of electricity usage during a billing period, typically measured in kilowatts (kW) [15]. It can be observed that demand charges are a significant portion of peaking in summer (e.g. \$3,863.51 in July). Meanwhile energy charges are the largest component, with the highest value in July (\$9,055.16).

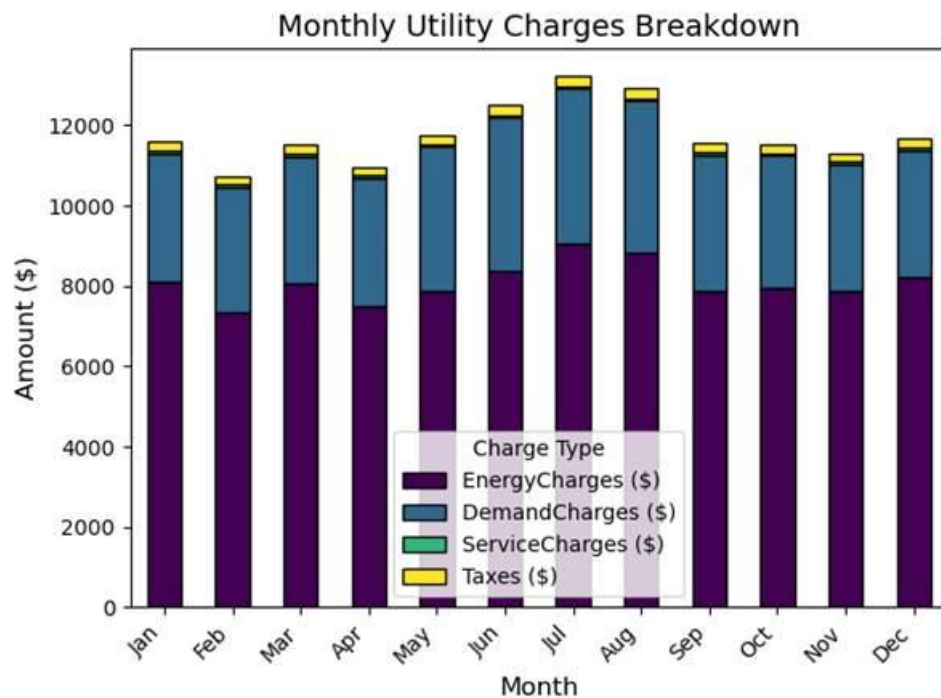


Figure 20; Monthly utility charge breakdown for electricity.

Figure 21 presents the monthly breakdown of gas charges, which suggests that during winter months the gas charges are higher, in line with the findings that gas consumption is mostly geared to meet the building's heating needs.

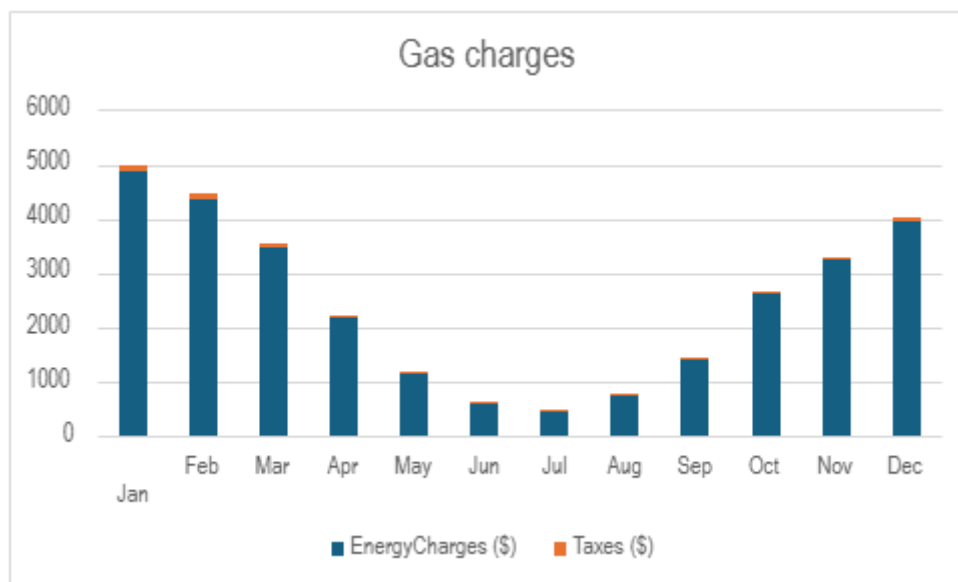


Figure 21; Monthly utility charge breakdown for gas.

Step 5: Improvement Proposal

Battery Energy Storage Optimization Plan

As shown in Figure 20, there is a large amount of cost spent on the demand charge, which is billed by peak power consumption each month. To address this, a battery energy storage system (BESS) is proposed to reduce the peak power consumption of this supermarket. The detailed analysis is shown in the following paragraphs, which incorporates electricity price inflation and evaluates investment returns standardized to 2025.

Table 17 shows the billing structure of the monthly electricity total cost. Take January as an example, the cost breakdown is shown in Table 18. The demand charge fee is ~30% of the total cost, which proves the necessity to reduce the peak power consumption.

Table 17; Electricity Billing Structure.

Charge Type	Calculation Formula	Rate	Unit	Remarks
Energy Charge	Total Monthly Consumption $\times$ Rate	\$0.0608	per kWh	Flat rate, no time-of-use
Demand Charge	Monthly Max 30-min Avg. Demand $\times$ Rate	\$11.65	per kW per month	Based on 30-minute interval peak
Monthly Service Fee	Fixed monthly fee	\$58.85	per month	Charged regardless of consumption
Tax	$2\% \times (\text{Energy} + \text{Demand} + \text{Service})$	2%	—	Local tax added to subtotal
Total Monthly Bill	Subtotal + Tax	—	—	Subtotal includes all fixed + variable items

Table 18; January 2010 Billing Breakdown.

Item	Value	Calculation	Amount (\$)
Monthly Energy Use	133,526 kWh	$133,526 \times 0.0608$	8,118.36
Monthly Peak Demand	272.37 kW	272.37×11.65	3,173.15
Monthly Service Fee	—	Fixed	58.85
Subtotal	—	$8,118.36 + 3,173.15 + 58.85$	11,350.37
Tax (2%)	—	$11,350.37 \times 0.02$	227.01
Total Bill	—	$11,350.37 + 227.01$	11,577.38

Table 19 shows the peak power consumption (peak demand) of each month, it can be observed that the annual peak demand is approximately 332 kW in July and typical monthly peaks range between 272-332 kW.

Table 19; Peak Demand of Each Month.

Month	Date & Time of Peak	Peak Demand (kW)
January	01/20 18:15	272.37
February	02/11 18:15	282.71
March	03/20 07:45	278.78
April	04/16 06:30	285.99
May	05/16 06:30	304.15
June	06/15 06:15	316.06
July	07/15 05:15	332.09
August	08/15 06:15	325.90
September	09/15 06:15	312.29
October	10/15 06:15	301.51
November	11/16 06:45	292.22
December	12/15 06:45	276.77

Considering the demand charge is a flat \$11.65 /kW-month with no blocks or penalties above any particular threshold, the savings grow linearly with Δ kW, hence the objective of the optimum battery size design is to get the highest savings per unit investment. For example, if the battery size is too small, the O&M and auxiliaries cost will take a large percentage that is less cost-effective, while if the battery size is too large, the diminishing marginal benefit will also reduce the savings and extend the payback time.

Consequently, the selection of battery size is 100 kW (80 kW usable, considering 80% maximum discharge depth), which can cut all the months' peak demand lower than 250 kW. The reason why choosing this size is that 100 kW is a standard hardware block and mass-produced in the market; the larger size may need more cabinets to accommodate it, and the smaller size will have a lower saving ratio.

Table 20 shows the detailed component selection of a 100 kW BESS system; the total investment cost is ~43k USD in 2025. Besides, to calculate the savings of demand power, it should also be standardized to 2025. As searched, the US CPI of electricity is changing from 193.095 to 283.688 [16] (2010-2025). So that the unit price of demand power will be 17.12 \$/kWh. Therefore, the below data/assumptions will be used for economic calculations (last 10 years):

- One-time CAPEX = \$ 43,000 (paid in 2025)
- Demand-charge rate in 2025 = \$17.12 /kW·month

- Peak-shaving block = 80 kW (battery PCS 100 kW, energy 215 kWh)
- Annual O&M = 2 % CAPEX = \$ 860 (assume fixed)
- Tariff inflation = 3 % / year
- Discount rate (real) = 4.5 %

Table 20; BESS Components Selection.

Item	Vendor /model	2025 Unit Price (\$)	Product Link/Remark
100 kW BESS	PREMIERPRO DIGITAL TECHNOLOGY SDN BHD	27,300	100kw Ess 215kwh Battery All In One Industrial & Commercial Outdoor Cabinet Energy Storage System Bess - Buy 100kw Industrial Energy Storage System 215kwh Outdoor Energy Storage Cabinet Bess Energy Storage System Product on Alibaba.com [17]
Bidirectional PCS – 100 kW @ 480 V	ATESS	5,560	Ateess PCS100 Bidirectional Battery Hybrid Inverter 100kw PV Solar Power Inverter with Inbuilt Transformer - Ateess Solar Inverter and Hybrid Inverter [18]
EMS & Site Controller	Acumen EMS™	3,000 (estimated)	AcumenEMSControlCenterAllInOneEnclosure.pdf [19]
Installation, Commissioning & Compliance Docs & Freight	\	7,172	20% of the equipment (estimated)
Total		43,032	The Real Cost of Commercial Battery Energy Storage in 2025: What You Need to Know GSL Energy [20] This link shows that the total cost for a 100kW BESS system will be between 25,000 and 50,000, which is reasonable

Table 21; 10 years discounted savings.

Year	Demand-charge rate (\$/kW·mo)	Gross Saving (\$)	O&M (\$)	Net Saving (\$)	Discount Factor (4.5 %)	Present Value (\$)
2025	17.12	16 435.2	860	15 575.2	0.9569	14 904.5
2026	17.63	16 924.8	860	16 064.8	0.9157	14 711.0
2027	18.16	17 433.6	860	16 573.6	0.8763	14 523.4
2028	18.71	17 961.6	860	17 101.6	0.8386	14 340.7
2029	19.27	18 499.2	860	17 639.2	0.8025	14 154.6
2030	19.85	19 056.0	860	18 196.0	0.7679	13 972.6
2031	20.44	19 622.4	860	18 762.4	0.7348	13 787.1
2032	21.06	20 217.6	860	19 357.6	0.7032	13 612.0
2033	21.69	20 822.4	860	19 962.4	0.6729	13 432.8
2034	22.34	21 446.4	860	20 586.4	0.6439	13 256.2
Discounted Savings						140,695

Regarding each parameter calculation, the formula is shown below:

$$DF_t = \frac{1}{(1+r)^t} (\text{DiscountFactor})$$

$$PV_t = \text{NetSaving}_t \times DF_t (\text{PresentValueofNetSaving})$$

$$\text{TotalPV} = \sum_{t=1}^{10} PV_t (\text{CumulativeDiscountedSavings})$$

$$NPV = \sum_{t=1}^{10} PV_t - \text{CAPEX} (\text{NetPresentValue})$$

$$NPV(r) = -\text{CAPEX} + \sum_{t=1}^{10} \frac{\text{NetSaving}_t}{(1+r)^t} = 0 (\text{Solvefor 'r' in IRR})$$

$$\text{Payback}_t = \text{NetSaving}_{\text{Year}_1} + \dots + \text{NetSaving}_{\text{Year}_t} \approx \text{CAPEX}$$

Table 22; NPV, IRR, and Payback Calculation.

Metric	Value
Net Present Value (NPV)	\$ 97,695 (140,695 - 43,000)
Internal Rate of Return (IRR)	~37 %
Payback Time	~3 years

Table 21 and Table 22 show the calculation of the economic factors of this investment in a 10-year horizon. The results show that the proposed 100 kW/215 kWh battery energy

storage system is highly cost-effective. Over a 10-year period, it is projected to generate discounted savings of approximately \$140,695, resulting in a net present value (NPV) of \$97,695. The system achieves a strong internal rate of return (IRR) of ~37% and pays back the initial investment in just under 3 years. These results confirm the system's excellent economic performance and justify its deployment as a robust solution for peak demand charge reduction.

Heat pump replacement of gas-based HVAC system

With over 50% of total annual energy consumption, the HVAC system is by far the largest energy consumption factor for the facility analysed. This application consumes 975 GJ of electricity and 6911 GJ of gas, at a cost of 23974\$ for electricity and 28680 \$ for gas, accounting for 30.8% of total annual costs for the supermarket. If the share of costs stemming from the HVAC system are considerable, so is the potential for improvement. With exergetic efficiencies in the range of 4.44% to 5.95%, the current system makes an inefficient use of the high chemical exergy of gas, burning at temperatures over 2000C° [21][22] to warm up air to a meagre 40C°. The thermoeconomic analysis further confirms these findings, suggesting that fuel costs from methane combustion account for 99.8% of the costs for the heating systems.

To reduce the costs associated with the room heating and cooling systems, namely the HVAC system, implementing modern heat pump devices in place of the current gas burner heating system is analysed as a cost-effective solution to reduce costs and emissions associated with the climatization of the supermarket. In this case, a simpler air source heat pump (ASHP) design would be selected, due to the simplified installation procedure, and the complexity of using the alternative ground source heat, since in Alaska the ground is frozen much of the year, which would complicate installation and maintenance. The costs of ASHP systems in the United States are around 463\$/kWth [23], which include equipment and installation costs, and are some of the lowest costs worldwide, whilst ground source heat pumps (GSHP) are between two to four times more expensive [23]. Consequently, ASHP are deemed the optimal solution for the reference building and location.

With a closed system like the one considered, using inlet air at 21C°, the International Renewable Energy Agency (IRENA) [23] estimates that for an outlet temperature of 35C°, close to the 40C° of our supermarket design outlet T°, the COP of a heat pump is around 5. The closed loop design would be maintained, as it increases the COP of heat pumps significantly; the study by IRENA [23] suggests that if the inlet temperature was -40C° as is the outside air, the COP would drop significantly to values below 2.

In the reference day, using the established air volume flows and a temperature jump from 21C° to 40C°, which is needed in the design day, the heating power required can be calculated as

$$W_{thermal_{out}} = \dot{m} * cp * \Delta T$$

Whilst the design day is already considered the limit of operation for the system, given that it is one of the most demanding days for heating needs, the resulting thermal power required will be multiplied by a factor of 1.5. This system oversizing is done in case it is needed to operate the heat pump in particularly adverse conditions. In particular, the system is designed for closed loop operation, warming air from 21C° to 40C°, but if the building gets much colder during winter, and gets near the outer temperatures which are usually much lower than 21C°, the COP can drop below 5, and a much greater thermal power will be needed to warm up the building.

The resulting value for the heat pump rated power is provided in Table 23 below, and this design load is multiplied by the cost factor of 463\$/kWth provided by IRENA, such that a CAPEX estimate can be obtained for the new heat pumps.

Table 23; Design point thermal loads for heat pump replacement of existing HVAC systems across all 6 zones of the supermarket.

Zone	PSZ-AC number	User design air flow rate (m3/s)	Required thermal output power (kW) with a 50% oversize factor	Heat pump equipment and installation costs (USD)
Office	PSZ-AC 1	0.749	27.78	12862.14
Dry storage	PSZ-AC 2	4.047	149.985	69443.055
Deli	PSZ-AC 3	1.410	52.215	24175.545
Sales	PSZ-AC 4	12.257	454.08	210239.04
Produce	PSZ-AC 5	4.178	154.815	71679.345
Bakery	PSZ-AC 6	1.466	54.3	25140.9

Assuming a COP of 5, as suggested by IRENA for parameters such as those of the design day considered previously in this study [23], the power requirements for each heat pump can be calculated as the desired thermal output, divided by the COP value of 5. Considering the reference heating day, energy savings amount to approximately 85%, as seen below in Table 24. However, the cost savings are not so large when contrasting the new operational cost with that of the traditional gas system, which uses both a gas heat exchanger and an electric fan to provide heating through the HVAC system. The new all electric system only saves 27% of costs for the reference day, and this savings percentage can be applied to the yearly gas and electricity consumption charges associated with the heating system. This amounts to 14374 \$ when extrapolated over the course of a year for the entire facility, insufficient to recover the investment costs of the heat pumps.

Table 24; Energy and cost savings for heat pump replacement of existing HVAC systems across all 6 zones of the supermarket.

Zone	Total energy savings (%)	Operational cost heat pump (\$/h)	Operational cost existing solution (\$/h)	Cost savings (%)
Office	83.79	0.327881784	0.404326247	18.91
Dry storage	84.61	1.770242958	2.500162935	29.19
Deli	84.31	0.616283202	0.792366714	22.22
Sales	84.59	5.359415424	7.398231423	27.56
Produce	84.64	1.827250482	2.584851657	29.31
Bakery	83.96	0.64089204	0.820823694	21.92
Total	84.5	10.54196589	14.50076267	27.30

Whilst the new heat pump proposed is much more energy efficient, the large difference in prices between gas and electricity largely nullify the economic benefit of the new equipment, given that gas costs 14.94\$/MWh against 88.52 \$/MWh for electricity, according to the reference data. As suggested in the literature [24], if the electricity price to gas ratio is higher than 3, gas furnaces are more economic than heat pumps, as is the case for this study. The current study considers electricity prices that are 5.92 times the gas prices on a kWh basis, which explains why the equipment replacement is not profitable in this location.

With an estimated lifetime of 25 years for heat pumps, as suggested by several sources and used in academic literature studies [25] [26] [27] an assessment can be made of the net present value of the investment over its life cycle. The maintenance costs for a gas boiler and a heat pump can be assumed roughly the same [28] [29] such that the payback calculations can exclude this factor. Assuming a discount rate of 4.5 % [30] and an inflation rate of 2.3% [31], the NPV calculations along the heat pump's lifetime show that the additional investment of installing the heat pump are never recovered through the lower utility costs, as illustrated in Figure 22 below.

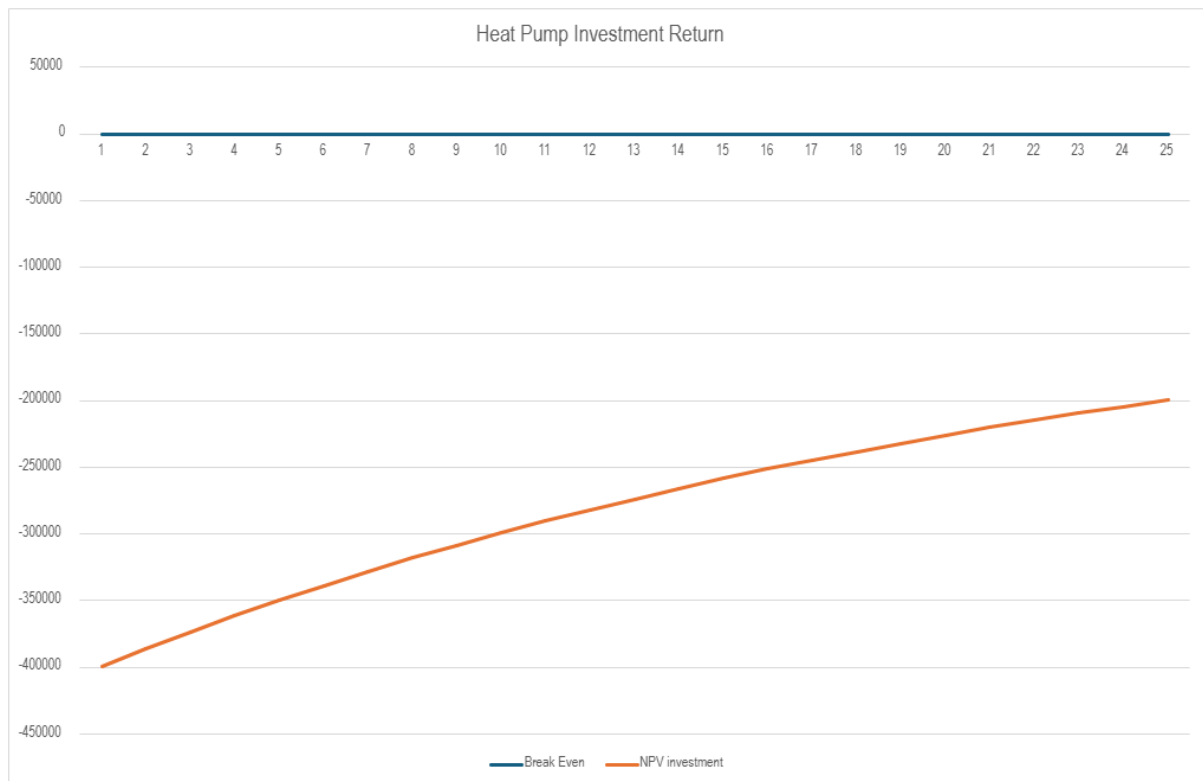


Figure 22; NPV of a heat pump replacement of the current HVAC system in the supermarket, under the current gas and electricity prices, showing that the investment does not break even over the 25 year lifespan.

Considering that the gas price to electricity price difference is lower in other parts of the country, a scenario with higher gas prices can be analysed, to determine the price difference that would render heat pump replacement of current gas heaters a profitable investment. The average electricity price in the United States goes from 10x gas prices to $\frac{3}{4}$ of gas price [24]. If gas prices doubled in the study location, the utility price spread would reduce to 2.96, as shown in Figure 23. In this new scenario, the investment breaks even after 21 years, which is too long to be considered a safe investment. The price spread reduces further to 1.97 if gas prices tripled, or to 1.48 if gas prices quadrupled. These new scenarios are illustrated in Figures 24 and 25, respectively, where the break-even period is lowered to 15 and 13 years for each scenario. Whilst these scenarios show a better business case, the increase of gas prices by a factor of three or four, whilst maintaining constant electricity prices, is not a realistic scenario.

Consequently, upgrading the heating system to heat pumps does not provide a positive economic case unless utility prices change drastically. However, if the location wished to reduce the overall energy consumption, due to environmental reasons or others, heat pumps can still be an attractive solution if higher costs can be accepted.

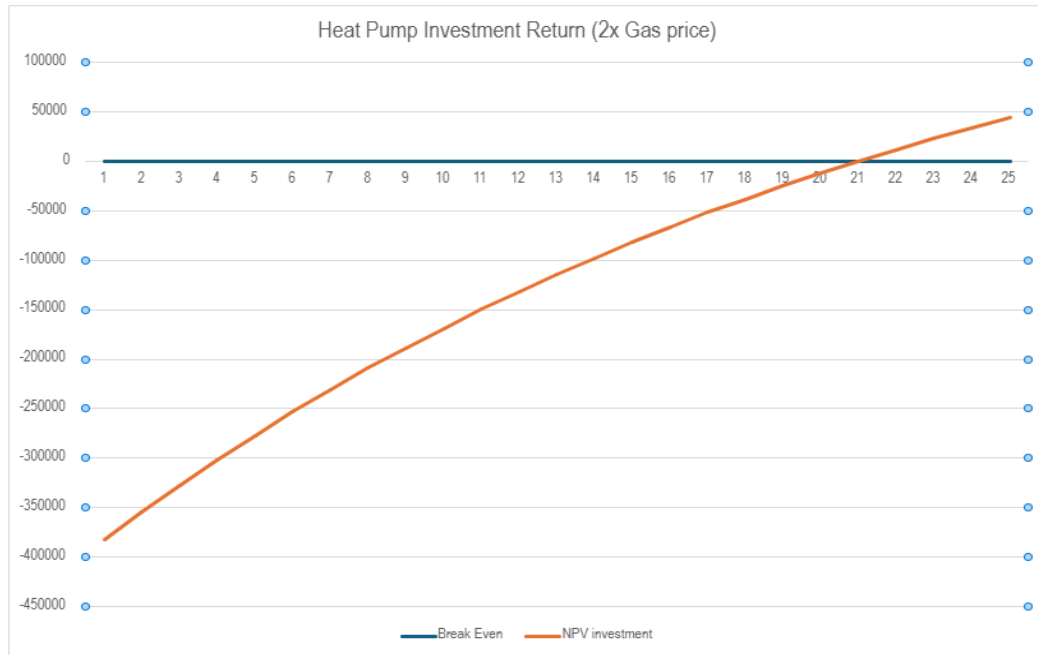


Figure 23; NPV of a heat pump replacement of the current HVAC system in the supermarket, under gas prices twice as high as current costs.

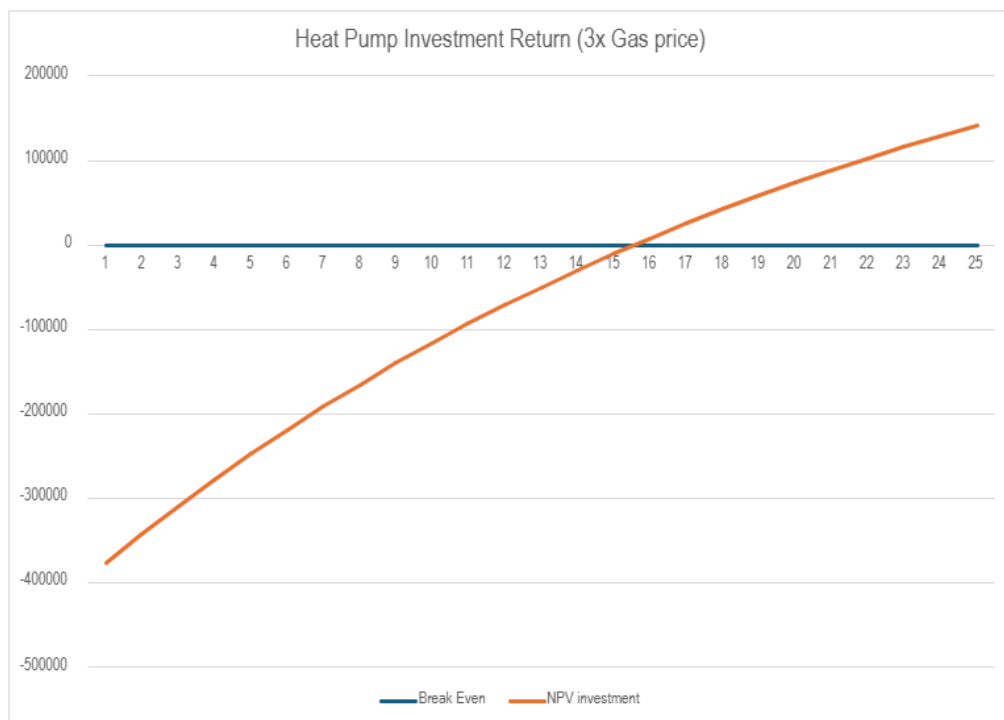


Figure 24; NPV of a heat pump replacement of the current HVAC system in the supermarket, under gas prices three times the current costs.

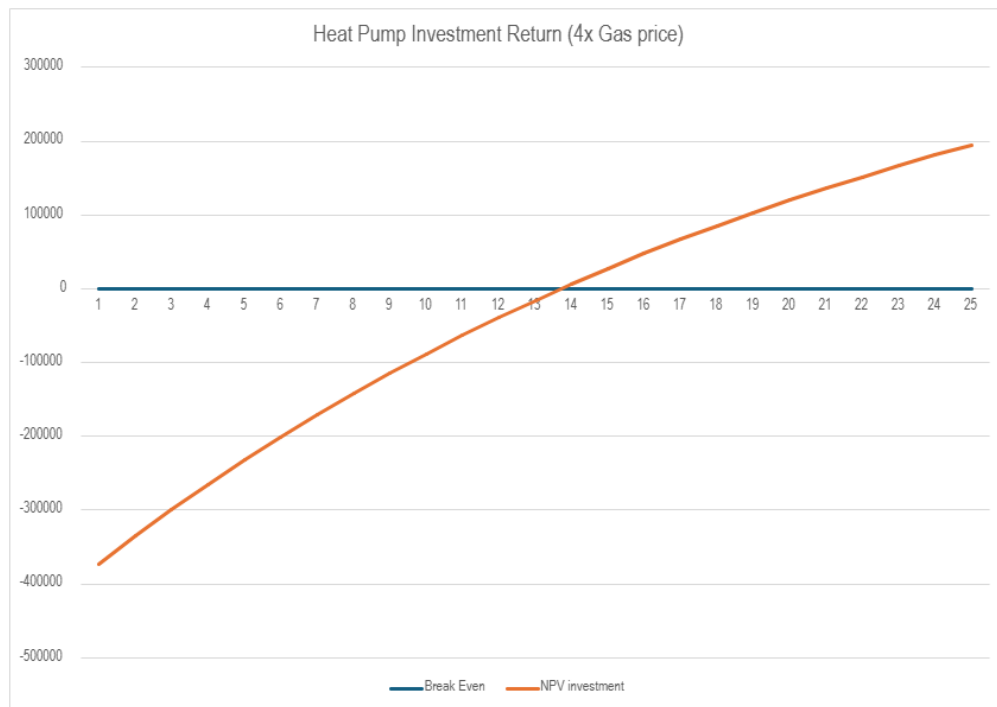


Figure 25; NPV of a heat pump replacement of the current HVAC system in the supermarket, under gas prices four times the current costs.

Envelope improvement

The improvement related to the envelope of the supermarket represents one of the largest opportunities for energy efficiency enhancement in the building. Analysis shows that 95.3% of total gas consumption is attributed to thermal losses through the building envelope, indicating that nearly all the energy used to heat the air is ultimately lost in the envelope. Furthermore, the current U-values of the envelope components, walls: 0.454 W/m²·K, windows: 2.615 W/m²·K, roof: 0.269 W/m²·K, and foundation: 3.120 W/m²·K, are significantly above the limits recommended for cold climates, highlighting a substantial margin for improvement. The heat losses for each of the four elements can be calculated using the following equation:

$$Q = A * U * \Delta T$$

The total annual energy loss can be determined by referencing the average temperatures of the outdoor air and soil, as reported in Table 25.

Table 25; Average Temperatures of air and soil.

Month	Average T	Average Soil T
January	-21	-3
February	-17	-1.8
March	-11	2
April	1.5	4.4
May	10.5	7
June	16	8.9
July	17	9.5
August	14	9.1
September	8	7.1
October	-1.5	4.5
November	-15	1.8
December	-19.5	-1.2

With this data, the total annual thermal energy losses through the building envelope can be accurately calculated.

Table 26; kW lost for each envelope segment.

Month	Foundation	Wall	Window	Roof	Total
January	313	31	19	47	410
February	297	28	17	43	385
March	248	24	15	35	322
April	217	14	9	22	262
May	182	8	5	12	207
June	158	4	2	6	168
July	150	3	2	4	159
August	155	5	3	8	171
September	181	9	6	15	211
October	215	17	10	25	267
November	250	26	17	40	333
December	289	30	18	46	363

As previously established, it is evident that the greatest share of thermal losses in the building originates from the foundation insulation.

Envelope Upgrade

Fairbanks is located in IECC Climate Zone 8, one of the coldest zones in the United States. Accordingly, the Alaska Building Energy Efficiency Standards (BEES) requires stringent insulation requirements to ensure thermal performance in extreme climates [32] The prescribed maximum U-values are as follows:

- Walls: $\leq 0.042 \text{ W/m}^2\cdot\text{K}$
- Roof: $\leq 0.020 \text{ W/m}^2\cdot\text{K}$
- Basement/Foundation: $\leq 0.050 \text{ W/m}^2\cdot\text{K}$
- Fenestration (windows): $\leq 0.300 \text{ W/m}^2\cdot\text{K}$

The upgraded envelope will therefore be constructed using improved materials that achieve U-values in compliance with these regulatory thresholds, as shown in Table 27.

Table 27; U-values for the four envelope segments.

Component	Material	Old U-value	New U-value
Roof	Fiberglass	0.269	0.02
Wall	Fiberglass batt plus continuous rigid foam	0.454	0.042
Foundation	Extruded polystyrene	3.120	0.05
Window	Vinyl low-E double pane	2.615	0.3

With these revised U-values, it becomes possible to calculate the updated thermal losses through the envelope components.

Table 28; kW lost for each part with the new improvements on the envelope.

Month	Foundation	Wall	Window	Roof	Total	Reduction factor (%)
January	5	2.8	2.2	3.6	13.6	97
February	4.8	2.6	1.9	3.2	12.5	97
March	3.9	2.1	1.7	2.6	10.5	97
April	3.4	1.3	1.0	1.6	7.4	97
May	2.9	0.7	0.5	0.88	5.1	98
June	2.5	0.3	0.3	0.4	3.5	98
July	2.4	0.2	0.2	0.4	3.2	98
August	2.5	0.5	0.3	0.58	3.9	97
September	2.9	0.87	0.68	1.1	5.6	98
October	3.5	1.5	1.1	1.8	8	97
November	4.0	2.4	1.8	3	11.3	97
December	4.6	2.7	2.1	3.4	12.9	96

The reduction in thermal losses is significant, with an average decrease of approximately 95% across all envelope components. The cost analysis has been performed assuming a 30-year service life, as supported by industry data [33]. The investment costs associated with each envelope upgrade are as follows.

Table 29; Investment costs for each part and their relative impact on the losses and on the price.

Component	Area (m ²)	Cost (\$/m ²)	Total Cost (\$)	Impact on the losses (%)	Impact on the price (%)
Roof	4181	16 [34]	66893	11.5	17
Wall	1610	14 [34]	22534	7.4	6
Window	175	1000 [35]	174700	4.6	45
Foundation	4181	30 [36]	125424	76.3	32

To estimate the energy savings enabled by the retrofit, it is assumed that 95.3% of total gas consumption is attributable to losses through the envelope, and that 97% of those losses can be mitigated through the installation of the new envelope system. The savings have been obtained knowing:

- the cost for MJ of natural gas: up to 0.00415 \$/MJ, according to the dataset
- The original heating power consumption throughout the year: 6911105 MJ

With these values the annual savings can be calculated as it follows:

$$\text{Annual Savings} = 6911105 * \text{CostperMJ} * (0.953) * (0.97)$$

A discount rate of 4.5% [30] and an inflation rate of 2.3% [31] has been applied in the financial evaluation. The resulting Net Present Value (NPV) calculations are presented in Figure 26.

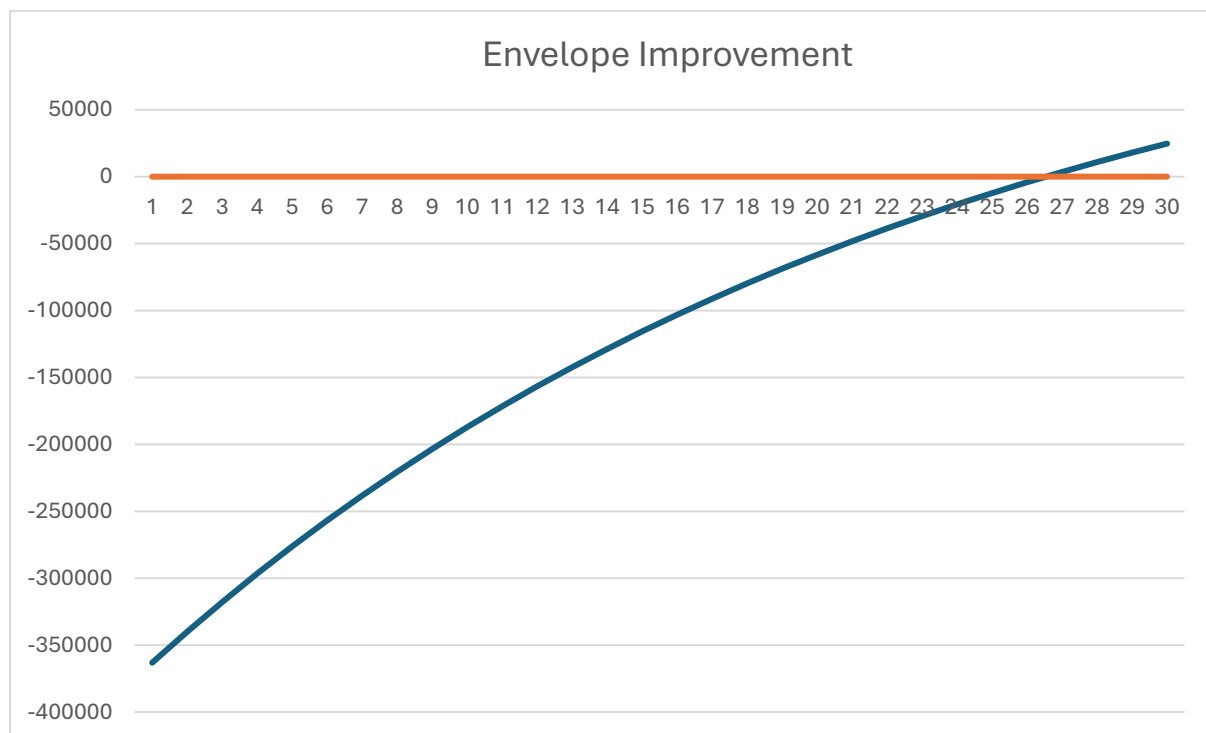


Figure 26; ROI with all the improvements made.

As shown, the breakeven point for the full-envelope retrofit is only reached in year 26, with relatively modest net gains over the system's lifetime. However, Table 29 highlights that while the impact of window upgrades on thermal losses is minimal, their associated costs are disproportionately high. By excluding the window retrofit and recalculating the investment performance, the Return on Investment (ROI) profile shifts significantly, as illustrated in Figure 27. In this revised scenario, the breakeven point occurs prior to year 11, and the cumulative savings by year 30 amount to approximately \$200,000.

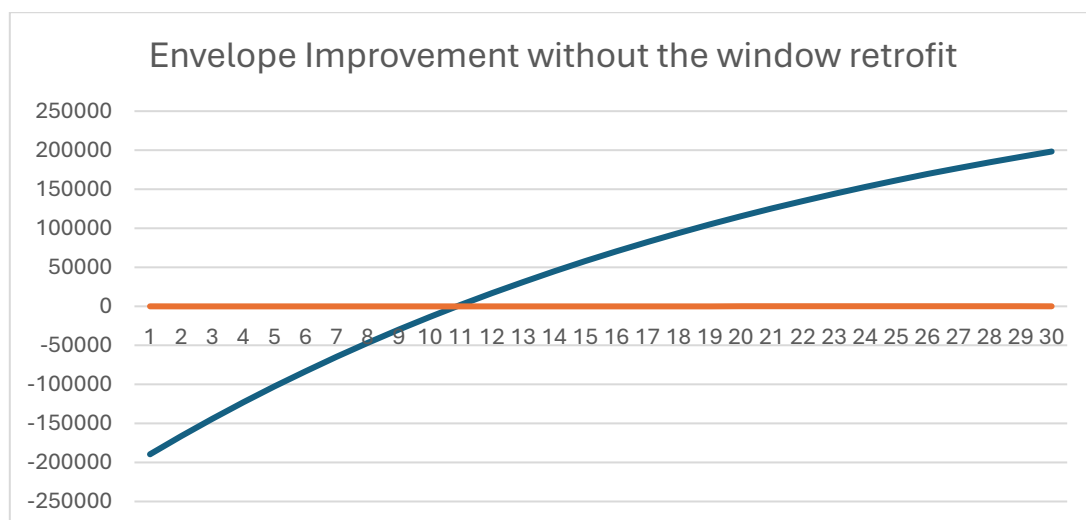


Figure 27; ROI without the window retrofit.

To conclude, it is clear that the improvements to the building envelope, particularly those targeting the roof, foundation, and walls, can result in a substantial reduction in energy consumption, making the retrofit not only technically effective but also financially viable under optimized conditions. While the window upgrades represent a costly investment with limited thermal return, the exclusion of this element dramatically improves the economic performance of the project, shortening the payback period and significantly increasing long-term savings. In contrast to HVAC system replacements, which were shown to be economically unjustified due to the high electricity-to-gas cost ratio, envelope improvements directly tackle the source of inefficiency, heat loss, making them a more strategic and impactful investment for this facility in Fairbanks.

Lightning Optimization Plan

Lightning accounts for a big part of the building's consumptions. Particularly, interior lightning accounts for a total of 933 GJ (~259 MWh), while exterior lightning for 211.66 GJ (~59 MWh). Moreover, if comparing its consumptions to the area of the building, it can be seen that its electricity intensity is of 237.97 MJ/m² (~76 kWh/m²), resulting the most intensive utility, even more than the HVAC that states a value of 236.50 MJ/m² (~66 kWh/m²). Consequently, lightning is a key area for optimization. The analysis will consider only the interior lightning, since not enough data are provided to do a detailed study of the external area.

The building is a Department of Energy (DOE) reference buildings, which predominantly use fluorescent lighting (fluorescent type is the standard model type of lighting for DOE reference building model). Fluorescent lights are more efficient than traditional incandescent bulbs, using about 60–80% less energy and lasting significantly longer (around 20 years) [37]. However, they contain small amounts of toxic mercury and are less efficient and more fragile compared to newer technologies [38]. They can be replaced with LED (light-emitting diode) systems, which offer even greater energy efficiency, longer lifespan, no hazardous materials, and better performance in various applications.

Again, the different zones of the supermarket have an area in m² that it is shown in Table 30, together with the current utilized W/m² of lightning.

Table 30; Floor area and lighting needs of the different zones.

Zone	Area (m ²)	Lighting (W/m ²)
Office	89	11.84
Dry storage	622	8.61
Deli	225	18.30
Sales	2325	18.30
Produce	711	18.30
Bakery	209	18.30

A method that can be used to find the necessary number of lights to be placed in the supermarket is the lumen method (or total flux method), which gives an output in lumens, that is the unit of measurement of the amount of light emitted by a light source.

$$Total\ flux\ (lumen) = \frac{Area\ (m^2) * Illuminance\ (lux)}{Utilization\ factor * Maintenance\ factor}$$

Guidelines related to the specific lighting of supermarkets suggest a general lightning of 30-50 footcandles (~300-500 lux) and a recommended light levels for task areas are 50–200 footcandles (~500-2000 lux) [39]. A value of 500 lux is taken as standard, since all the considered areas are frequented by the public and minor lights adjustments for specific zones could be achieved knowing where are needed. Furthermore, the

utilization factor considers the reflection index (r) and the dimensions of the area (k), and as a standard value is taken 0.7; meanwhile, the maintenance factor is chosen based on the frequency of the cleaning and maintenance operation. For a supermarket, a value of 0.85 is considered acceptable.

6. Coefficiente di pulizia M:

Dipende dalla frequenza degli interventi di manutenzione e pulizia.

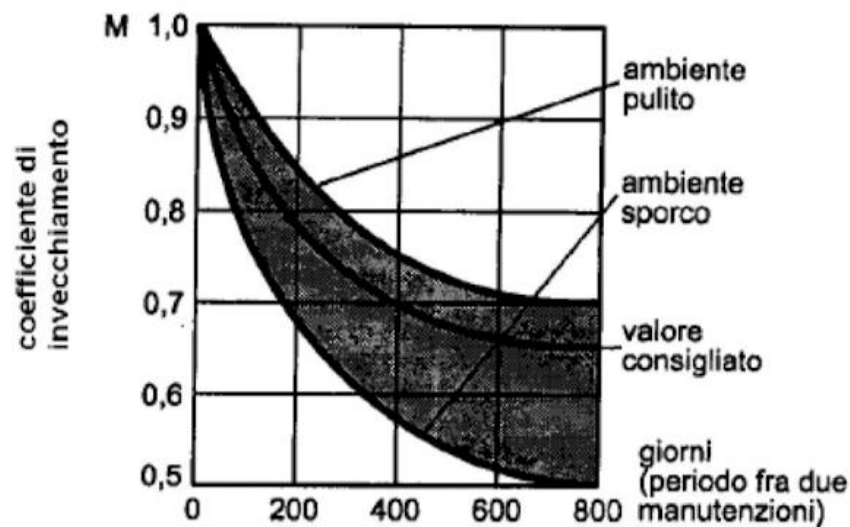


Figure 28; Maintenance factor values scheme [40].

Having all the data, it is possible to calculate the total flux values for each area of the facility, in Table 31. Subsequently, an architectural linear LED light is chosen as a reference [41], with an average lumens output of 3000 lumens and a price of roughly 120€ per unit. This type of light is specifically designed for supermarkets. It is thus possible to calculate the number of lights needed per each area.

Table 31; Total Flux for the six areas.

Zone	Total flux (lumen)	Lights number
Office	74,790	25
Dry storage	522,689	174
Deli	189,076	63
Sales	1,953,782	651
Produce	597,479	199
Bakery	175,630	59

This type of lights uses a maximum of 25 W per unit. For a total of 1,171 lights, it results in a power consumption of 29.3 kW.

Figure 2 shows the lightning schedule per weekday, Saturday and Sunday, from which it is possible to estimate a total consumption of 330.85 kWh per singular weekday, 289.86 kWh per Saturday and 169.82 kWh per Sunday. In a yearly basis these values accounts for a consumption of 110.25 MWh, more than a half less than the precedent 259 MWh consumptions (just interior lightning) and showing the possibility of a ROI study.

Table 32; Cost of the improvements.

	Unit	LED lights	Fluorescent light
Investment cost (year 0)	€	140,538	0
Annual electricity consumption	MWh/year	110.25	259
Interest rate	%	4.5%	4.5%
Electricity Price (same for 10 years)	€/kWh	0.19	0.19

As shown in the graph above, a ROI study enhances that the investment is recovered in a period of slightly less than 6 years.

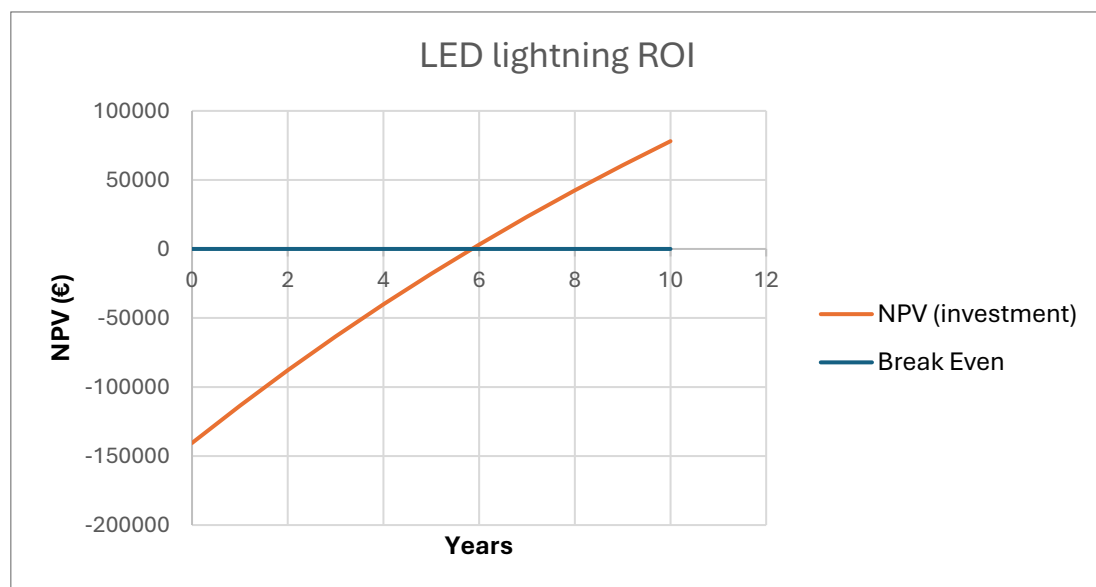


Figure 29; ROI of the LED.

Finally, if the location wished to reduce the lightning electricity consumption, and at the same time improve the lightning performances, LEDs are a good option if a return time of investment of 6 years is retained acceptable. Not enough information related to the actual fluorescent lighting is provided (remaining lifetime, when were installed) to assess a detailed comparison. To conclude, improving the lightning reflects in a substantial

reduction in energy consumption and financially viable operation with a break-even period of 6 years and cumulative savings of roughly 200,000€ over the approximately 20 years system's lifetime.

Refrigeration Optimization Plan

By far, the largest electricity consumption is the refrigeration, which accounts for 2824.22 GJ (~784.5 MWh), and including all the systems focused on the maintenance of goods' temperature. Even, if not enough data is provided to achieve a detailed energy/exergy audit, some interesting insights can be found.

As already know, the supermarket is located in Fairbanks (Alaska), with winter peak temperatures of approximately -26.1°C to -31.7°C , although it's common for temperatures to reach -40.0°C , or even -45.6°C ; during summer, the hottest month in Fairbanks is July, with an average high of about 22.2°C and an average low of about 12.2°C [42]. These data show how, particularly in winter, it could be interesting to consider the outside air, which is at a very low temperature, to refrigerate the goods.

This kind of system is generally known as “free cooling” (or an air-side economizer) and has been studied extensively in cold-climate contexts, even in the rising cooling-demanded sector of data centers. Free cooling is a refrigeration strategy that utilizes naturally cold outdoor air, rather than mechanically compressing refrigerant, to remove heat from a system and lowering its energy consumptions [44] [45].

In the supermarket case, a free-cooling setup would circulate a secondary cooling fluid (such as glycol or chilled water) through an outdoor “economizer” heat exchanger whenever winter temperatures drop below the refrigeration setpoint; the outdoor air then chills the fluid, which in turn cools your display cases, bypassing the compressor need [44] [45]. By integrating dampers, mixing boxes and pre-heat coils to prevent humidity, the system prevents coil freeze-up even when temperatures dip to -40°C [46], together with controls rated for extreme cold operation [46].

A smart control system switches between free cooling, partial compressor use and full mechanical refrigeration, allowing savings on cooling energy during Alaska's cold winters, maintaining the products' quality [44] [45].

Conclusion

The energy efficiency project of a Reference Supermarket in Fairbanks, Alaska, identifies significant opportunities for improving the energy efficiency and reducing the operational costs. The site analysis found that 55.4% of total energy consumption was devoted to natural gas, primarily for heating, whereas the remaining 44.6% of consumption stemmed from electricity, mostly used for refrigeration systems, fans, interior equipment and lighting. However, a cost analysis reveals that electricity represents the largest annual cost at 141,241.79\$ against the 29,638.78\$ of gas consumption, resulting from the price difference between both commodities.

By conducting a detailed energy audit for the most relevant equipment, the key areas for optimization are found to be the building envelope, refrigeration appliances, peak electricity demand charges and the lighting system. Furthermore, HVAC systems are also key areas of improvement, with a specific focus on heating operations which constitute most of the operation time for the selected location, and is the activity with the largest energy consumption.

Improvement suggestions include adding a 100kW, 215 kWh BESS system, which reduces peak power consumption, and can provide an IRR of 37%. An upgrade of the building envelope is suggested to reduce heat losses, by changing the insulation on the roof, walls and floor, to obtain a ROI of \$200,000 over the investment's lifetime. Replacing current lights with LED lighting is also analysed, which can provide a ROI of over \$88,584.29 (78,064.31€) over a ten years lifespan. Further improvements are discussed, such as utilizing outside air for refrigeration needs, using free cooling to reduce energy demand. Replacing the gas-based HVAC with heat pumps proved less feasible due to the high electricity-to-gas price ratio. Results suggest that whilst heat pumps drastically reduce energy consumption, the replacement of gas by electricity is not economically profitable for the current utility prices, and therefore this improvement is not recommended.

All in all, substantial energy and cost savings can be obtained from the suggested upgrades of building envelope, lighting, refrigeration, and the installation of a BESS system to reduce peak demand charges. The payback time of suggested investments ranges from 3 to 11 years, proving their economic value, their ability to reduce costs for the supermarket owners, and improve their environmental footprint through a reduced energy usage.

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Appendix

Table 1.1; Office's exergy calculation values

OFFICE			
OBJECT	VALUE	unit of measure	data source
Max Flow Fan Rate	0.75	m ³ /s	File data
Ratio kgH ₂ O/kgAir	0.00006		File data
Flow rate water	0.000058185	kg/s	C
Flow rate air	0.96975	kg/s	C
Pressure drop fan	622	Pa	File data
P _{outlet} air	100300	Pa	C
T _{outlet} air	40	C°	File data
T _{outlet} air	313.15	K	C
Physical exergy AIR out	1.116526157	kJ/kg	C
Physical exergy AIR out	1.082816206	kJ/s	C
Fan Power	0.8536	kW	File data
Design Load	17.598	kW	File data
Nominal Efficiency	0.8		File data
Power before combustor	21.9975	kW	C
Flow rate gas	0.00043995	Kg/s	C
Chemical exergy GAS	22.61643926	kJ/s	C
Physical exergy GAS	0.168038903	KJ/s	C
Exergy efficiency	0.045808132		C

Table 1.2; Dry storage's exergy calculation values

DRY STORAGE			
OBJECT	VALUE	unit of measure	data source
Max Flow Fan Rate	4.05	m ³ /s	File data
Ratio kgH ₂ O/kgAir	0.00006		File data
Flow rate water	0.000314199	kg/s	C
Flow rate air	5.23665	kg/s	C
Pressure drop fan	1109.65	Pa	File data
P _{outlet} air	100787.65	Pa	C
T _{outlet} air	40	C°	File data
T _{outlet} air	313.15	K	C
Physical exergy AIR out	1.525979505	kJ/kg	C
Physical exergy AIR out	7.991500037	kJ/s	C
Fan Power	7.591	kW	File data
Design Load	95.423	kW	File data
Nominal Efficiency	0.78		File data
Power before combustor	122.3371795	kW	C
Flow rate gas	0.002446744	Kg/s	C

Chemical exergy GAS	125.7793562	kJ/s	C
Physical exergy GAS	0.934533714	KJ/s	C
Exergy efficiency	0.059502674		C

Table 1.3; Deli's exergy calculation values

DELI			
OBJECT	VALUE	unit of measure	data source
Max Flow Fan Rate	1.41	m ³ /s	File data
Ratio kgH2O/kgAir	0.00006		File data
Flow rate water	0.000109388	kg/s	C
Flow rate air	1.82313	kg/s	C
Pressure drop fan	622	Pa	File data
P_outlet air	100300	Pa	C
T_outlet air	40	C°	File data
T_outlet air	313.15	K	C
Physical exergy AIR out	1.116526157	kJ/kg	C
Physical exergy AIR out	2.035694468	kJ/s	C
Fan Power	1.758	kW	File data
Design Load	33.239	kW	File data
Nominal Efficiency	0.78		File data
Power before combustor	42.61410256	kW	C
Flow rate gas	0.000852282	Kg/s	C
Chemical exergy GAS	43.81312705	kJ/s	C
Physical exergy GAS	0.325529129	KJ/s	C
Exergy efficiency	0.044353873		C

Table 1.4; Sale's exergy calculation values

SALES			
OBJECT	VALUE	unit of measure	data source
Max Flow Fan Rate	12.26	m ³ /s	File data
Ratio kgH2O/kgAir	0.00006		File data
Flow rate water	0.000951131	kg/s	C
Flow rate air	15.85218	kg/s	C
Pressure drop fan	1017.59	Pa	File data
P_outlet air	100695.59	Pa	C
T_outlet air	40	C°	File data
T_outlet air	313.15	K	C
Physical exergy AIR out	1.44883361	kJ/kg	C
Physical exergy AIR out	22.96854921	kJ/s	C
Fan Power	20.767	kW	File data
Design Load	290.207	kW	File data
Nominal Efficiency	0.78		File data

Power before combustor	372.0602564	kW	C
Flow rate gas	0.007441205	Kg/s	C
Chemical exergy GAS	382.5288414	kJ/s	C
Physical exergy GAS	2.842168299	KJ/s	C
Exergy efficiency	0.056553557		C

Table 1.5; Produce's exergy calculation values

PRODUCE			
OBJECT	VALUE	unit of measure	data source
Max Flow Fan Rate	4.18	m ³ /s	File data
Ratio kgH ₂ O/kgAir	0.00006		File data
Flow rate water	0.000324284	kg/s	C
Flow rate air	5.40474	kg/s	C
Pressure drop fan	1109.65	Pa	File data
P _{outlet} air	100787.65	Pa	C
T _{outlet} air	40	C°	File data
T _{outlet} air	313.15	K	C
Physical exergy AIR out	1.525979505	kJ/kg	C
Physical exergy AIR out	8.248017322	kJ/s	C
Fan Power	7.837	kW	File data
Design Load	98.709	kW	File data
Nominal Efficiency	0.78		File data
Power before combustor	126.55	kW	C
Flow rate gas	0.002531	Kg/s	C
Chemical exergy GAS	130.110712	kJ/s	C
Physical exergy GAS	0.96671545	KJ/s	C
Exergy efficiency	0.059374807		C

Table 1.6; Bakery's exergy calculation values

BAKERY			
OBJECT	VALUE	unit of measure	data source
Max Flow Fan Rate	1.466	m ³ /s	File data
Ratio kgH ₂ O/kgAir	0.00006		File data
Flow rate water	0.000113732	kg/s	C
Flow rate air	1.895538	kg/s	C
Pressure drop fan	622	Pa	File data
P _{outlet} air	100300	Pa	C
T _{outlet} air	40	C°	File data
T _{outlet} air	313.15	K	C
Physical exergy AIR out	1.116526157	kJ/kg	C
Physical exergy AIR out	2.116544745	kJ/s	C
Fan Power	1.99	kW	File data

Design Load	34.515	kW	File data
Nominal Efficiency	0.8		File data
Power before combustor	43.14375	kW	C
Flow rate gas	0.000862875	Kg/s	C
Chemical exergy GAS	44.35767707	kJ/s	C
Physical exergy GAS	0.329575106	KJ/s	C
Exergy efficiency	0.045344245		C